

University of Sheffield

Deep-Learning-Based Segmentation of Stratum Corneum in Ultra-High-resolution OCT Images of Living Skin

Author: Eljilani Ahmed Hbal -

Supervisor: Professor Stephen Matcher

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1 Abstract

This project investigates the application of deep learning for the automated segmentation of the Stratum Corneum (SC) in ultra-high resolution Optical Coherence Tomography (OCT) images of living skin. Accurate SC segmentation is crucial for dermatological research. As the skin's primary barrier, its thickness and impermeability are critical indicators for assessing skin conditions, cosmetic efficacy, and pharmaceutical permeation. However, accurate delineation of the SC remains challenging due to its thin structure, weak intensity gradients, and the presence of speckle noise inherent to OCT imaging. Traditional manual segmentation methods are time-consuming, subject to inter-observer variability, and difficult to scale across large clinical datasets, highlighting the need for robust and reproducible automated methods.

Three deep learning models were trained on a dataset of 212 manually labelled OCT images, expanded to 1,272 via data augmentation. Pixel-level annotation masks were paired and validated against source images using filename-based matching. Consistent preprocessing was applied consistently across all models, including Contrast Limited Adaptive Histogram Equalisation (CLAHE) and ImageNet normalisation. Segmentation was performed across four classes: Background, SC, Skin Surface (SS), and the Dermo-Epidermal Junction (DEJ).

Three deep learning models were trained and evaluated under a controlled experimental framework to ensure a fair architectural comparison: U-Net, a widely adopted encoder-decoder network for biomedical image segmentation; DeepLabV3+, a model employing atrous convolution and Atrous Spatial Pyramid Pooling (ASPP) for multi-scale contextual feature extraction; and U-Net++, an extended encoder-decoder model using redesigned skip connections and dense feature aggregation to improve the segmentation of fine skin structures.

U-Net established a strong baseline with a Mean IoU of 0.8488 and Dice coefficient of 0.9136, demonstrating reliable detection of the SC and surrounding epidermal layers. DeepLabV3+ attained a Mean IoU of 0.7929 and Dice coefficient of 0.8740, showing effective broad anatomical localisation but reduced boundary precision at the SC interface due to its limited reliance on high-resolution spatial skip connections. U-Net++ achieved the strongest overall performance across all metrics, with a Mean IoU of 0.8860, Dice coefficient

of 0.9372, Sensitivity of 0.9670, and Precision of 0.9108; these results demonstrate that its dense skip connections and nested encoder-decoder structure provide meaningful improvements in SC boundary delineation compared to the other evaluated architectures.

These findings suggest that automated deep learning frameworks can effectively overcome the inherent limitations of OCT imaging, providing a robust tool for the objective assessment of skin barrier function in clinical and pharmaceutical research.

2 Individual Contribution

My individual contribution to this project is the design and implementation of a deep-learning-based segmentation pipeline for OCT skin images including dataset preprocessing, the development and training of multiple segmentation models, and the quantitative and qualitative analysis of results. The existing resources shown below were used to assist the execution of this project.

Software:

- Dataset Labeling: OCTlabelTool (V1), a MATLAB based image labeling tool developed by Dr. Robert Byers, University of Sheffield, used for manual labeling of skin layer boundaries in the OCT dataset.
- Deep learning Framework: PyTorch, for model development and training
- Segmentation Library: segmentation-models-pytorch, for encoder backbone integration
- Augmentation Library: Albumentations, for data augmentation during training
- Image Processing: OpenCV, for preprocessing and CLAHE application

Data:

- OCT Dataset: Raw OCT B-scans of human skin provided by Dr. Revin, obtained through correspondence facilitated by Prof. Stephen J. Matcher, University of Sheffield, used for training, validation, and testing of all models.

Hardware:

- GPU: NVIDIA T4, accessed via Google Colaboratory, used for all model training and experimentation

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4 Introduction

4.1 Background and Motivation

Ultra-high resolution Optical Coherence Tomography (OCT) is a non-invasive imaging technique that produces cross-sectional images of living skin at micrometre-level resolution. Unlike traditional biopsy-based methods, OCT allows the internal structure of the skin to be examined in real time without causing damage to the tissue. This makes it particularly valuable in dermatological research, where repeated and non-destructive measurements are often required. OCT images reveal the layered structure of the epidermis, including structures such as the Stratum Corneum (SC), Skin Surface (SS), and the Dermo-Epidermal Junction (DEJ), each of which plays a distinct biological role.

The Stratum Corneum is the outermost layer of the skin and serves as the body's main barrier against environmental threats, transepidermal water loss, and the penetration of external substances. Its thickness and structural integrity are clinically important indicators used in the assessment of skin conditions such as eczema, psoriasis, and ichthyosis, where barrier dysfunction is a defining feature. Beyond clinical diagnosis, accurate SC thickness measurement has a compelling application in the cosmetic and pharmaceutical industries. The SC acts as the main barrier to topical product absorption, meaning its thickness and condition directly determine how effectively active ingredients in creams, serums, and therapeutic formulations penetrate the skin. By providing precise and automated measurements of SC thickness from OCT images, deep learning segmentation tools could enable researchers to objectively evaluate and compare the efficacy of different formulations across individuals and body sites, without the need for invasive testing. This has direct implications for the development of more targeted, evidence-based skincare and transdermal drug delivery products.

However, precisely segmenting the SC in OCT images is challenging due to its extremely thin profile, typically spanning only a few pixels in thickness, combined with weak intensity gradients, speckle noise inherent to coherence-based imaging, and natural variation in thickness across individuals and anatomical sites. On palmoplantar skin the SC appears as a well-defined superficial band, whereas on non-palmoplantar skin it may span only a few micrometres with poorly-defined boundaries, making consistent automated detection across diverse imaging conditions a significant computational challenge.

Manual segmentation of OCT images is not a practical solution at scale. It is time-consuming, requires significant clinical expertise, and is prone to inter-observer variability, meaning different annotators may produce inconsistent results for the same image. These limitations motivate the development of automated segmentation methods that can produce reliable, reproducible, and efficient analyses of SC structure across large datasets.

Deep learning has emerged as the leading approach for automated medical image segmentation, owing to its ability to learn complex spatial features directly from labelled data. This project evaluates three state-of-the-art models specifically chosen for their potential to address the challenges of OCT imaging: U-Net, for its efficient use of spatial skip connections; DeepLabV3+, for its multi-scale contextual feature extraction; and U-Net++, which employs dense, nested skip connections to improve the refinement of thin, low-contrast structures. By comparing these models, this study aims to identify the most robust framework for automated SC delineation.

4.2 Aims and Objectives

4.2.1 Aims

The primary goal is to implement and evaluate deep learning architectures for the precise automated segmentation of the Stratum Corneum (SC) and surrounding epidermal layers in ultra-high resolution OCT images. This research seeks to identify which architectural characteristics best address the challenges of thin-layer segmentation, assessing the suitability of each model for clinical and pharmaceutical applications including non-invasive thickness measurement and barrier function assessment.

4.2.2 Basic Objectives:

- Conduct a literature review on deep learning methods for OCT image segmentation and similar biomedical imaging applications to understand current challenges.
- Study and analyse existing models, including U-Net, DeepLabV3+, and U-Net++, focusing on their design, advantages, and limitations in medical image analysis.
- Preprocess and label the provided OCT image dataset.
- Implement and train deep learning models on the labelled dataset using training, validation, and testing splits.
- Evaluate and compare model performance using classification based metrics.

- Interpret and discuss the results in terms of segmentation quality.

4.2.3 Advanced Objectives:

- Fine-tune existing deep learning models by adjusting hyperparameters, training strategies, and data augmentation methods.
- Test different training settings to ensure model convergence and to identify which approach produces the most accurate and consistent segmentation results.
- Critically evaluate clinical limitations, discussing how the optimised models could support dermatological research or diagnostic use.

4.3 Project Planning and Management

4.3.1 Gantt Chart

The project was organised into distinct milestones, encompassing basic and advanced objectives scheduled from the start of Semester 1 through to the final submission. Progress consistently aligned with the planned timeline, with several key milestones achieved ahead of the projections outlined in the Gantt chart. Initial phases focused on aim definition, risk assessment, and the configuration of the development environment. This structured approach facilitated consistent progress tracking and allowed for necessary adjustments when specific tasks required more intensive development than originally anticipated.

The Gantt chart served as a primary management tool throughout the project to monitor weekly tasks and evaluate progress. A comparison between the projected and actual timelines indicates that dataset labelling, model architecture configuration, and preliminary training runs were completed within the scheduled periods. The chart below illustrates the full project lifecycle and the successful completion of all scheduled tasks.

Eljilani Ahmed - Advanced project

Project start: **Wed, 4/1/2026**

Display week: **1**

TASK	ASSIGNED TO	PROGRESS	START	END
Deadlines and Milestones				
Aims and Objectives	Eljilani Ahmed	100%	10/3/25	10/20/25
Interim report	Eljilani Ahmed	100%	10/20/25	12/8/25
Interview with Second Reader	Eljilani Ahmed	100%	12/8/25	12/16/25
Dissertation	Eljilani Ahmed	100%	2/9/26	5/11/26
Oral Presentation	Eljilani Ahmed	50%	5/11/26	5/20/26
Project planning				
Gantt Chart	Eljilani Ahmed	100%	10/3/25	10/20/25
Lab Diary	Eljilani Ahmed	100%	10/3/25	10/3/25
Risk Assessment	Eljilani Ahmed	100%	10/3/25	10/20/25
Brainstorming & Initial Ideas	Eljilani Ahmed	100%	10/3/25	10/10/25
Techincal work (Basic Objectives)				
Literature review on deep learning for OCT	Eljilani Ahmed	100%	11/8/25	12/8/25
Study and Analyse U-Net, DeepLabV3+, and U-Net++	Eljilani Ahmed	100%	10/3/25	10/24/25
Preprocess and Label Dataset	Eljilani Ahmed	100%	10/23/25	3/31/26
Implement and Train Models	Eljilani Ahmed	100%	11/10/25	4/22/26
Evaluate and Compare Model Performance	Eljilani Ahmed	100%	2/11/26	5/5/26
Interpret and Discuss Initial Results	Eljilani Ahmed	100%	11/24/25	12/5/25
Interpret and Discuss Final Results	Eljilani Ahmed	100%	4/24/26	5/8/26
Techincal work (Advanced Objectives)				
Fine Tune and Optimise Models	Eljilani Ahmed	100%	4/10/26	4/22/26
Test Different Training Settings	Eljilani Ahmed	100%	4/22/26	5/2/26
Clinical Relevance and Discussion	Eljilani Ahmed	100%	4/11/26	5/6/26

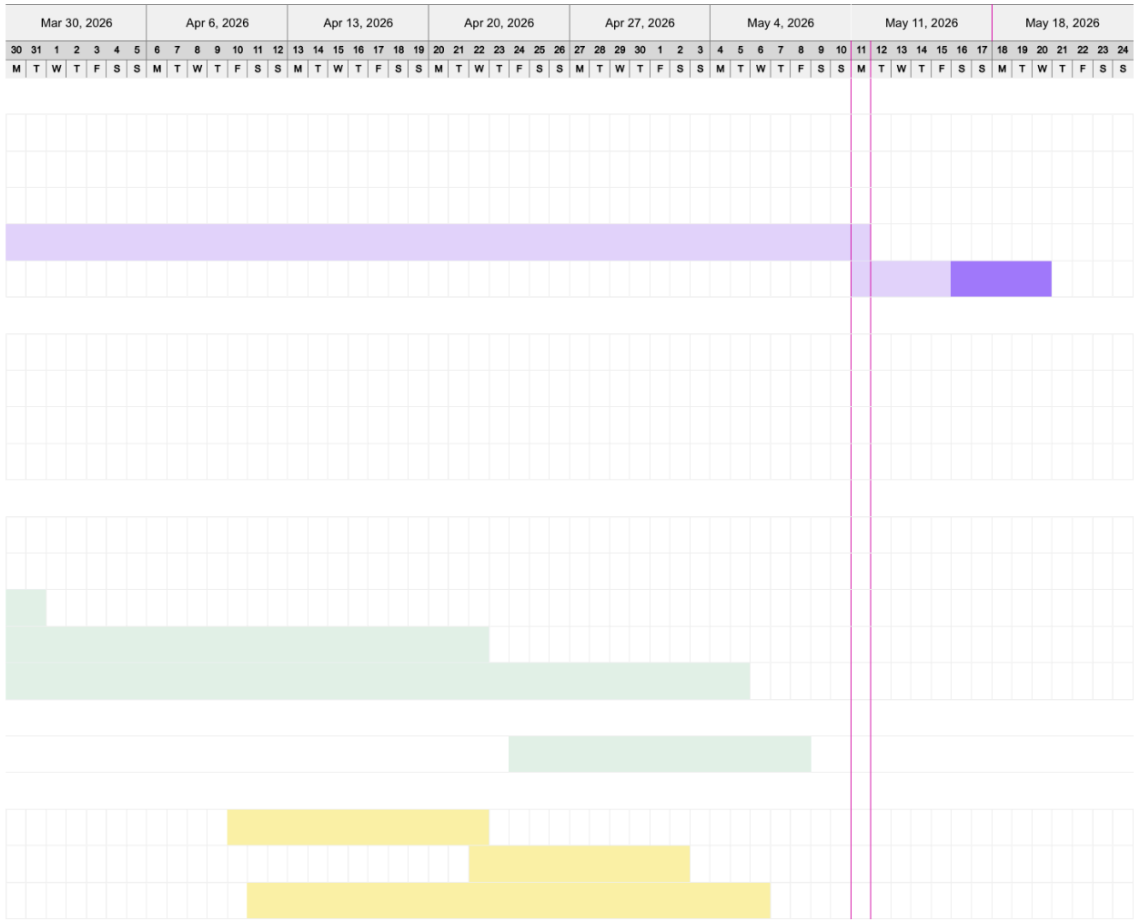


Figure 1: Gantt Chart

4.3.2 Risk Management

Risk management was an important step to ensure the successful completion of this project within the allocated timeframe. The risks associated with this project fell into two categories: data security risks and technical project risks. Each identified risk was assessed alongside a corresponding mitigation strategy.

Risk	Mitigation
Unauthorised access to OCT images	Files were stored exclusively on password-protected university computers or encrypted hardware with two-factor authentication.
Accidental storage on unprotected devices	File paths were verified prior to saving to ensure data remained within protected university directories.
Loss of data due to device failure	Redundancy was maintained through dual-site storage: one encrypted physical copy and one university server copy.
Retention of data beyond project end	After the project ends, delete the dataset from the USB drive and university computer.
Ethical compliance in outputs	Ethical approval was secured prior to the inclusion of data in any academic reports or presentations.

Table 1: Data Security Risks

Risk	Mitigation
Model training failure due to GPU memory limitations	Batch sizes and image resolutions were adjusted; gradient accumulation was used where necessary.
Google Colab session timeout during long training runs	Model checkpoints were saved frequently to allow training to resume from the last stable state.
Overfitting due to small dataset size	Applied heavy data augmentation, early

	stopping, and dropout regularisation to improve generalisation.
Poor annotation quality affecting model performance	Implemented a sequential labelling strategy with manual visual inspection and correction of all generated masks.
Time constraints preventing completion of advanced objectives	Basic objectives were prioritised to ensure a core comparative study was delivered before moving to advanced experiments.

Table 2: Technical Project Risks

This project is low risk because it uses a pre-existing dataset with no contact with participants. All work will follow the university's data protection guidance. The images will only be used for academic analysis and will be stored in protected locations. No raw data will be shared outside the project. All files will be deleted at the end of the dissertation period. These steps ensure secure handling of the dataset and full compliance with the ethical standards required for this work.

4.3.2 Overview of Report

This report is organised into five primary chapters, each reflecting a key stage in the development and evaluation of the deep learning segmentation pipeline:

- **Chapter 4** introduces the project, providing background and motivation for automated SC segmentation in OCT imaging, along with the aims, objectives, and project planning.
- **Chapter 5** presents a comprehensive literature review covering OCT imaging and skin structure, noise and artefacts in OCT images, traditional segmentation methods, the mathematical foundations of deep learning, and a detailed review of the three architectures evaluated in this project — U-Net, DeepLabV3+, and U-Net++. The chapter concludes with a comparative analysis of the three architectures and an identification of the research gap motivating this work.
- **Chapter 6** shows the methodology and system design, starting on how the dataset was labelled, followed by the performance and computational metrics including Pixel Accuracy, Intersection over Union (IoU), Dice Coefficient, Sensitivity (Recall), Specificity, Precision, Parameters, GFLOPs, and Latency. After the metrics, the chapter continues and concludes by explaining the data preparation and training configuration of the models.

- **Chapter 7** focuses on the implementation, training, and evaluation of each model: U-Net (baseline), DeepLabV3+ (multi-scale contextual model), and U-Net++ (dense feature aggregation). Quantitative, class-wise, and qualitative analyses are provided for each, followed by a rigorous cross-model comparison, and concluded by clinical recommendation.
- **Chapter 8** discusses the broader implications of the results, including the conclusions drawn from the project, directions for future work, and sustainability and ethical considerations.

5 Literature review

5.1 OCT Images and Skin Structure

Optical Coherence Tomography (OCT) is a non-invasive imaging modality that produces high-resolution, cross-sectional images of biological tissues based on the principles of low-coherence interferometry [1]. In dermatology, it has emerged as an important tool for *in vivo* skin imaging, offering a distinctive balance of resolution and penetration depth that positions it between high-frequency ultrasound and reflectance confocal microscopy [2]. OCT allows for the visualisation of subsurface skin structures to a depth of 2 mm with a resolution of approximately 10 μm, enabling visualisation of important features including the epidermis and dermo-epidermal junction without invasive tissue excision [3].

In OCT imaging, contrast arises from spatial variations in the optical scattering properties of different tissue components, thus enabling the differentiation of subsurface skin structures including the epidermis, dermo-epidermal junction, and dermis [1,3]. However, the delineation of adjacent tissue boundaries is often limited when neighbouring layers exhibit similar optical properties, reducing boundary contrast and hindering precise segmentation [1]. A further inherent limitation is the presence of speckle noise—a granular interference pattern that arises from the constructive and destructive interference of backscattered light from randomly distributed tissue scatterers—which obscures fine structural detail and further complicates the segmentation of tissue boundaries in OCT images [7,8].

Human skin is a multi-layered organ consisting primarily of the epidermis, dermis, and subcutaneous tissue. The outermost sublayer of the epidermis, the stratum corneum (SC), is a thin but functionally important structure responsible for barrier protection, regulation of

transepidermal water loss, and defence against external environmental factors [5]. In OCT imaging, the visibility of the SC is highly site-dependent; on palmoplantar skin it can be identified as a well-defined, homogeneous, low-scattering superficial layer due to its greater thickness at these sites; conversely, on non-palmoplantar skin the SC is generally not visible at conventional OCT resolution [1]. However, the ultra-high-resolution OCT system used in this project operates beyond conventional resolution limits, improving the visibility of thin epidermal structures such as the SC and motivating its application to this segmentation task.

Accurate identification and segmentation of the SC in OCT images remain challenging due to its thinness, weak contrast with neighbouring layers, and variability across different anatomical regions [1,6]. Furthermore, even micrometre-scale variations in SC thickness can substantially affect barrier integrity and the penetration of therapeutic agents, making precise and reliable segmentation a key objective in dermatological analysis and disease assessment [6].

A thorough understanding of both the structural makeup of the skin and its representation in OCT images is therefore essential, providing the foundation for the development of the deep-learning-based segmentation methods evaluated in this project.

5.2 Noise and Artifacts of OCT Images

Optical Coherence Tomography (OCT) images are inherently prone to multiple sources of noise and imaging artefacts that degrade image quality [7]. The most pervasive source of this degradation is speckle noise. As a coherence-based imaging modality, OCT is subject to the constructive and destructive interference of backscattered and forward-scattered photons from randomly distributed tissue scatterers [4,8]. This phenomenon manifests as a granular, multiplicative noise pattern that fundamentally limits both the effective resolution and the contrast-to-noise ratio of the acquired tomograms. Unlike additive sensor noise, speckle is signal-dependent and obscures fine morphological details, smoothing out the already subtle gradients between adjacent sublayers, such as the boundary between the SC and the viable epidermis [8]. Consequently, the presence of speckle complicates traditional edge-detection algorithms, necessitating advanced software-based filtering or hardware-based compounding to recover lost structural information [9].

Beyond speckle, imaging the skin *in vivo* involves a high degree of refractive index mismatch at the air-tissue interface. The irregular structure of the skin, including dermatoglyphic patterns, causes the incident OCT beam to scatter [10]. This mismatch results in intensity artefacts, often appearing as vertical, low-signal streaks beneath skin grooves, as well as geometric distortions where deeper structures appear displaced due to local changes in the optical path length. These distortions disrupt the continuity of the epidermal layers, posing a distinct challenge for algorithms attempting to track the thin SC across a B-scan. Furthermore, attenuation artefacts arise when highly scattering or absorbing structures at or near the skin surface impede the penetration of light into deeper tissues. Features such as dense hair follicles, sebaceous glands, or hyperkeratotic scales can cast prominent shadows [1]. While the SC is less susceptible to being shadowed by deeper tissue, substantial variations in its own thickness can severely attenuate the signal, masking the underlying dermo-epidermal junction.

Finally, *in vivo* OCT acquisition is inherently sensitive to patient movement. Even minor involuntary motions, such as micro-tremors or respiration, can introduce significant spatial misalignments during the scanning process [11]. These motion artefacts typically manifest as jagged discontinuities within a single B-scan or across adjacent frames. For automated analysis, the confluence of speckle noise, optical distortions, shadowing, and motion artefacts ensures that raw OCT images present a highly noisy environment for feature extraction. These factors blur weak boundary contrast and compromise the structural continuity of the SC [6,12]. Therefore, modern approaches increasingly rely on robust deep learning architectures, which can be trained to recognise broader contextual and spatial features, effectively distinguishing true anatomical boundaries from coherent noise and imaging artefacts [12].

5.3 Traditional Segmentation Methods in OCT

Prior to the adoption of deep learning, segmentation of layered structures in OCT images was predominantly achieved through classical image processing and machine learning techniques. These methods can be broadly classified into edge-based detection, graph-based methods, and active contour models, each carrying unique assumptions about image structure that ultimately limit their applicability to the specific challenges of skin OCT.

Edge-based methods, including gradient operators such as Canny and Sobel filters, detect layer boundaries by identifying sharp intensity transitions in the image [13]. While computationally efficient, these methods are weak and not suited to skin OCT, where the SC boundary produces only weak and inconsistent intensity gradients due to its thinness and optical similarity to the underlying viable epidermis. Speckle noise further corrupts these gradients, causing edge-detection algorithms to produce fragmented or spurious boundaries that do not correspond to true anatomical structures [8]. Consequently, edge-based methods require extensive preprocessing, such as speckle-noise filtering, which itself risks smoothing out the very fine boundaries they are intended to detect—a fundamental contradiction that limits their reliability in this context.

Graph-based methods, most notably the Optimal Graph Search Segmentation Framework proposed by Li et al. [14], represent a more principled approach. By modelling the image as a weighted graph and formulating layer boundary detection as a minimum-cost path problem, these methods can simultaneously segment multiple surfaces while enforcing smoothness constraints [14]. This model has shown strong performance in retinal OCT segmentation, where layer boundaries are relatively well-defined and consistent across subjects [15]. However, its translation to skin OCT is considerably more problematic. The cost functions underpinning graph-search methods must be manually engineered to reflect expected intensity profiles and boundary characteristics, making them highly sensitive to the site-dependent variability of skin structure. The SC, which is clearly visible on palmoplantar skin but nearly unnoticeable at non-palmoplantar sites, violates the homogeneity assumptions these cost functions rely upon, reducing their generalisability across different imaging conditions [1].

Active contour models, or “snakes”, offer an alternative by iteratively deforming a curve toward image features such as edges or intensity gradients under the influence of internal smoothness constraints [16]. Level-set formulations extend this idea to handle topological changes, making them more flexible in theory [17]. In practice, however, active contours are highly sensitive to initialisation and prone to convergence on local minima, particularly in images with low contrast or high noise levels [16]. For the SC, where the boundary signal is weak and spatially variable, these methods frequently require manual initialisation or region-specific parameter tuning, undermining the goal of automated, reproducible segmentation.

A critical limitation shared across all traditional methods is their reliance on hand-crafted features and explicit assumptions about boundary appearance. This constrains their ability to generalise across the natural variability of skin type, anatomical site, and imaging condition in a realistic OCT dataset [18]. Moreover, none of these methods exploit the broader spatial context of the image—the recognition that the SC's location and thickness are partially predictable from the global organisation of surrounding epidermal layers. It is precisely this contextual reasoning, learned implicitly from labelled data, that gives deep learning models their key advantage in this segmentation task [12], motivating the architectural choices explored in this project.

5.4 Mathematical Foundations of Deep Learning

Deep learning models are a class of machine learning systems built from artificial neural networks with multiple layers of non-linear, learnable transformations. A rigorous understanding of the mathematical foundations and principles underpinning these models is essential for interpreting architectural design choices and evaluating why certain configurations perform better than others in tasks such as SC segmentation in OCT imaging.

At the core of the neural network is the artificial neuron, which computes a weighted sum of its inputs and passes the result through a non-linear activation function. For a neuron receiving an input vector *x*, with weight vector *w* and bias *b*, the output is given as:

$$a = f(w^T x + b)$$

Equation 1: Single Neuron's Output

where $f(.)$ denotes the activation function [19]. The choice of activation function is critical to the network's ability to learn complex mappings. The sigmoid function, defined as $\sigma(z) = \frac{1}{(1+e^{-z})}$ was historically dominant but suffers from the vanishing gradient problem in deep networks, where gradients decrease exponentially during backpropagation, effectively preventing early layers from learning [20]. The Rectified Linear Unit (ReLU), defined as $f(z) = max(0, z)$, largely alleviates this by maintaining a constant gradient for positive inputs, enabling the training of much deeper models and becoming the standard activation function in modern convolutional networks [21].

In convolutional neural networks (CNNs), the fundamental operation is the 2D discrete convolution. For a two-dimensional input feature map *I* and a learnable kernel *K* of size $m \times n$, the output feature map *S* is computed as:

$$S(i,i) = (I \times K)(i,j) = \sum_m \sum_n I(i+m, j+n) K(m,n)$$

Equation 2: 2D Discrete Convolution

This operation allows the network to detect local spatial patterns, such as edges and textures, in a translation-invariant manner [22]. Critically, the weights of *K* are shared across all spatial positions, dramatically reducing the number of learnable parameters compared to fully-connected layers and making CNNs particularly well-suited to image data [22]. In the context of OCT segmentation, successive convolutional layers learn progressively abstract representations, from low-level intensity gradients relevant to edge detection, to high-level structural patterns corresponding to tissue layer organisation.

Training a neural network involves minimising a loss function *L* that quantifies the discrepancy between the model's predictions and the ground-truth labels. For pixel-level segmentation tasks, the cross-entropy loss is commonly employed. For a binary segmentation problem with predicted probability $\hat{y}$ and ground truth label *y*, the binary cross-entropy is defined as:

$$L = -[y log(\hat{y}) + (1-y) log(1-\hat{y})]$$

Equation 3: Binary Cross Entropy

However, in medical image segmentation, class imbalance is a persistent challenge; the SC occupies only a small fraction of the total image pixels, meaning a model can achieve high accuracy simply by predicting the background everywhere [23]. To mitigate this, the Dice loss, derived from the Sørensen–Dice similarity coefficient, is frequently used either alone or in combination with cross-entropy:

$$L_{Dice} = 1 - (2\sum_i y_i \hat{y}_i)/(\sum_i y_i + \sum_i \hat{y}_i)$$

Equation 4: Dice Loss

This formulation directly penalises overlap failure between predicted and ground-truth regions, making it far more sensitive to the correct segmentation of small, localised structures such as the SC [23, 41, 43].

Parameters are updated iteratively through backpropagation, which applies the chain rule of calculus to compute the gradient of the loss with respect to each weight in the network [20]. The weight update at iteration t using stochastic gradient descent (SGD) is defined as:

$$w_{(t+1)} = w_t - n\nabla_{wl}$$

Equation 5: Stochastic Gradient Descent (SGD)

Where η is the learning rate, a critical hyperparameter controlling the magnitude of each update [19]. In practice, adaptive optimisers such as Adam, which maintain per-parameter learning rates adjusted by estimates of the first and second gradient moments, have largely superseded "vanilla" SGD in medical imaging applications due to their faster and more stable convergence [24].

A key challenge in training deep networks is overfitting, whereby the model learns to memorise training examples rather than generalise to unseen data. This is particularly acute in medical imaging where labelled datasets are typically small. Regularisation techniques such as dropout, which randomly deactivates neurons during training with probability p, and L_2 weight decay, which penalises large weight magnitudes, are standard countermeasures [19]. Batch normalisation, which normalises layer activations across a mini-batch, additionally stabilises training dynamics and has been shown to act as an implicit regulariser [25]. Collectively, these mathematical foundations — convolution, activation, loss formulation, and optimisation — underpin every architectural decision made in the segmentation models evaluated in this project.

5.5 Deep Learning for Biomedical and OCT Segmentation

The emergence of deep learning has significantly transformed the landscape of biomedical image segmentation, offering data-driven alternatives to the hand-crafted feature pipelines that characterised traditional approaches. Convolutional neural networks, in particular, have demonstrated a capacity to learn hierarchical representations of image structure directly from labelled examples, generalising across the natural variability of biological tissue in ways that

classical methods cannot [26]. This shift has been especially impactful in medical imaging, where the complexity and subtlety of anatomical boundaries demand robust, context-aware feature extraction.

The major contribution that established deep learning as the dominant paradigm in biomedical segmentation was the introduction of fully-convolutional networks (FCNs) by Long et al. [27]. By replacing the fully-connected layers of classification networks with convolutional layers, FCNs enabled dense, pixel-wise predictions across an entire image with a single forward pass. This design shift was critical because it preserved spatial information throughout the network, a prerequisite for precise boundary localisation. However, the repeated downsampling inherent in FCNs resulted in coarse output maps that lacked fine boundary detail—a limitation that subsequent models sought to address [27].

In the domain of OCT specifically, early deep learning approaches demonstrated clear superiority over traditional graph-based and active-contour methods. Fang et al. [28] applied a CNN-based framework to retinal OCT layer segmentation, demonstrating that learned features substantially outperformed manually engineered cost functions in both accuracy and robustness to image noise. Critically, this advantage was most pronounced precisely in the conditions most relevant to skin OCT—high speckle noise, low boundary contrast, and inter-subject variability—suggesting that the shift to deep learning is not merely a performance improvement but a qualitative change in how segmentation problems can be approached.

Subsequent work in skin-specific OCT segmentation has remained comparatively sparse relative to retinal imaging, where large annotated datasets such as the Duke DME dataset have accelerated architectural development [29]. This disparity reflects a broader challenge in dermatological OCT research: the absence of large, standardised, publicly available labelled datasets makes direct benchmarking difficult and limits the generalisability of reported results. Even when deep learning models are applied to skin OCT, performance varies considerably depending on anatomical site, imaging resolution, and the specific layers targeted, underscoring the need for task-specific model evaluation rather than the uncritical transfer of architectures validated on retinal data [30].

Data augmentation has emerged as a particularly important strategy in this context, compensating for the scarcity of labelled skin OCT images by artificially expanding the effective training set through geometric transformations, intensity perturbations, and elastic deformations [26]. While augmentation improves generalisation, it introduces its own design choices—transformations must preserve the biological plausibility of the augmented images, and inappropriate augmentation can introduce artefacts that mislead the model. This tension between augmentation breadth and biological validity is a recurring consideration in the training of segmentation models for thin-layer structures such as the SC.

Transfer learning offers a complementary strategy, initialising network weights from models pre-trained on large natural image datasets such as ImageNet before fine-tuning on the target medical imaging task [31]. Although the domain gap between natural images and OCT tomograms is substantial, pre-trained encoders have consistently been shown to provide better initialisation than random weights, accelerating convergence and improving performance particularly when labelled data is scarce [31]. However, the degree to which ImageNet features transfer to the specific textural and structural characteristics of skin OCT remains an open question, and represents one of the empirical considerations addressed in the model training described in this project.

Collectively, the literature establishes that deep learning is the most viable current approach to automated OCT segmentation, but that its application to skin—and to the SC specifically—remains an incompletely solved problem. The architectural choices, training strategies, and evaluation frameworks reviewed in the following sections are motivated by this gap.

5.6 U-Net for biomedical and OCT Segmentation

U-Net, introduced by Ronneberger et al. [32] in 2015, represents one of the most significant architectural contributions to biomedical image segmentation. Originally developed for the segmentation of neuronal structures in electron microscopy images, its design principles have proven remarkably generalisable across a wide range of medical imaging modalities, including OCT. Its enduring adoption in the field makes it a natural starting point for examining both its architectural foundations and its specific strengths and limitations in the context of skin layer segmentation.

The architecture follows a symmetric encoder-decoder structure. The encoder, or contracting path, applies successive blocks of convolutional layers followed by max-pooling operations, progressively reducing spatial resolution while increasing the number of feature channels. This process captures increasingly abstract semantic representations of the input image, encoding contextual information about tissue organisation at multiple spatial scales [32].

The decoder, or expansive path, mirrors this structure, using transposed convolutions to progressively restore spatial resolution and recover fine boundary detail. The defining innovation of U-Net is the introduction of skip connections, which concatenate feature maps from corresponding encoder layers directly to the decoder at each resolution level [32]. These connections allow the decoder to combine high-level semantic context from the bottleneck with low-level spatial detail from the encoder, preserving fine structural information that would otherwise be lost through repeated downsampling. This property is particularly consequential for SC segmentation, where the target layer occupies only a few pixels in thickness and its precise boundary must be recovered at full image resolution.

The performance of U-Net in biomedical segmentation has been extensively validated. Ronneberger et al. [32] demonstrated that the architecture could achieve competitive segmentation accuracy with as few as thirty labelled training images when combined with aggressive data augmentation—a finding of direct relevance to skin OCT research where large annotated datasets are rarely available. Subsequent studies have confirmed this data-efficiency advantage across diverse tasks, including retinal layer segmentation [33], cardiac MRI [34], and histological tissue classification [35], establishing U-Net as a robust baseline against which newer architectures are routinely benchmarked.

In OCT-specific applications, U-Net and its variants have demonstrated strong performance in retinal segmentation tasks. Roy et al. [33] introduced ReLayNet, a U-Net variant adapted for retinal OCT layer segmentation, incorporating a modified loss function combining cross-entropy and Dice loss to address class imbalance between thin retinal layers and surrounding tissue—a challenge directly analogous to the SC segmentation problem addressed in this project. Their results demonstrated that the skip connection mechanism was critical to recovering fine layer boundaries, with ablation experiments showing substantial

performance degradation when skip connections were removed [33]. This finding provides empirical justification for the architectural choice of U-Net as a primary model in this project.

However, U-Net is not without limitations. The standard architecture processes input images at a fixed resolution, which can constrain its ability to capture multi-scale contextual information simultaneously. In skin OCT, where the SC must be distinguished from adjacent layers that exhibit similar local texture, this single-scale processing can lead to boundary confusion in regions of low contrast [6]. Furthermore, the symmetric encoder-decoder structure results in a relatively large number of parameters, which increases the risk of overfitting when training data is limited and necessitates careful regularisation strategies. Several architectural extensions have been proposed to address these limitations, including U-Net++ [36], which introduces nested and dense skip connections to improve feature aggregation, and Attention U-Net [37], which incorporates attention gates to selectively suppress irrelevant background features during decoding. While these variants offer performance improvements in specific tasks, they also introduce additional complexity and hyperparameters, making systematic evaluation more challenging and is beyond the scope of this project.

For the purposes of this project, the standard U-Net architecture serves as the primary baseline model, providing a well-understood reference point against which the performance of DeepLabV3+ and U-Net++ can be meaningfully compared. Its established strengths in thin-layer segmentation and data-scarce environments make it a principled choice for the SC segmentation task, while its known limitations motivate the investigation of architecturally distinct alternatives.

5.7 DeepLabV3+ for biomedical and OCT Segmentation

DeepLabV3+, introduced by Chen et al. [38] in 2018, represents a distinct architectural approach to semantic segmentation that diverges fundamentally from the encoder-decoder paradigm of U-Net. Rather than recovering spatial resolution through a symmetric decoding pathway, DeeplabV3+ employs atrous convolution (also known as dilated convolution) as its central mechanism for capturing multi-scale contextual information while preserving feature map resolution [38].

Atrous convolution modifies the standard discrete convolution by introducing a dilation rate *r*, which inserts $r - 1$ zeros between consecutive kernel elements, effectively expanding the receptive field of the filter without increasing the number of learnable parameters or reducing spatial resolution through downsampling. Formally, for a one-dimensional signal (generalisable to 2D) *x* and a filter *w* of length *K,* the atrous convolution output *y* is defined as:

$$y[i] = \sum_{k=1}^{K} x[i + r \cdot k]\, w[k]$$

Equation 6: Atrous Convolution Output

This lets the network collect data over progressively larger spatial regions, so it can capture both fine detail and broad contextual structure within a single convolutional layer [38]. This is particularly important for OCT segmentation because the SC needs to be identified not only by its local intensity characteristics but also by its position in relation to other epidermal layers—a contextual signal that needs a sufficiently large receptive field to capture meaningful contextual signals.

The key difference of DeepLabV3+ over its predecessors is the Atrous Spatial Pyramid Pooling (ASPP) module, which applies multiple parallel atrous convolutions with different dilation rates to the same feature map, concatenating their outputs to produce a rich multi-scale representation [38]. By simultaneously processing the image at multiple effective scales, ASPP enables the model to handle the large amount of variation in SC thickness and appearance across different anatomical sites and individuals. The multi-scale processing capability represents a meaningful architectural advantage over standard U-Net, which operates at a fixed resolution within each encoder stage and relies on skip connections rather than explicit multi-scale feature aggregation to recover boundary detail.

In the context of skin OCT, the multi-scale feature aggregation offered by ASPP is especially valuable because the SC presents differently depending on anatomical site and individual variation. On palmoplantar skin it appears as a thick,well-defined superficial band, whereas on non-palmoplantar skin it may span only a few micrometres with poorly defined boundaries [1]. A fixed-scale model risks failing at one of these extremes, whereas ASPP's parallel dilation rates allow simultaneous sensitivity to both presentations of the same

structure. Furthermore, because ASPP preserves feature map resolution rather than discarding spatial detail through repeated pooling, boundary predictions are not degraded before the final output is produced, a meaningful advantage when the target layer occupies only a small number of pixels vertically [38].

DeepLabV3+ typically employs a ResNet or Xception backbone as its encoder, both of which benefit substantially from ImageNet pre-training [32,38]. In low-data availability settings such as skin OCT, this pre-trained network can accelerate convergence and improve generalisation, partially compensating for the limited availability of labelled training images. However, a notable drawback of DeepLabV3+ is its relatively weak decoder. Unlike U-Net, which uses a symmetric decoder with explicit skip connections to recover fine spatial detail from encoder feature maps at each resolution level, DeepLabV3+ relies primarily on simple bilinear upsampling after the ASPP module [38]. Without direct high-resolution feature pathways from encoder to output, fine boundary detail that is suppressed during encoding is harder to recover, which can produce slightly blurred or imprecise boundary localisation at full image resolution. For stratum corneum segmentation, where pixel-level precision is clinically meaningful [6], this represents a genuine architectural limitation relative to U-Net.

Computationally, DeepLabV3+ is more demanding than U-Net due to its deeper backbone and the ASPP module, requiring more memory and longer training times. This has practical implications when working with the ultra-high-resolution OCT images used in this project. Nevertheless, its strong multi-scale contextual reasoning makes it a well-motivated architectural choice for comparison, since it represents a fundamentally different segmentation strategy to U-Net—one that prioritises broad contextual understanding over explicit boundary recovery through skip connections.

5.8 U-Net++ for biomedical and OCT Segmentation

U-Net++, introduced by Zhou et al. [36] in 2018, was developed to address a fundamental limitation of the standard U-Net architecture: the semantic gap between encoder and decoder feature maps connected through direct skip connections. In the original U-Net, skip connections concatenate encoder feature maps to their corresponding decoder layers without any intermediate processing, meaning the decoder must bridge a potentially large representational gap between low-level spatial features and high-level semantic

representations. U-Net++ addresses this by replacing direct skip connections with dense nested convolutional blocks that progressively refine feature maps before they are passed to the decoder [36].

The key architectural innovation of U-Net++ is its nested encoder-decoder structure. Rather than a single linear encoder-decoder pathway, U-Net++ introduces a series of dense skip pathways that connect nodes at the same resolution level across multiple decoder depths. Each intermediate node aggregates feature maps from all preceding nodes at the same scale, enabling progressive feature refinement at every resolution level. This means that by the time features reach the final decoder, they have undergone multiple stages of refinement, substantially reducing the semantic gap that standard U-Net skip connections leave unaddressed [36]. This progressive refinement is particularly beneficial for segmenting thin and low-contrast structures such as the SC, where the precise boundary must be reconstructed from features that span a wide range of semantic abstraction levels.

A further advantage of U-Net++ is its support for deep supervision, whereby intermediate outputs at different decoder depths can be used to compute the loss during training [36]. This encourages the network to learn meaningful representations at multiple scales and can improve convergence, particularly when training data is limited. Additionally, deep supervision enables model pruning at inference time—shallower decoder paths can be used for faster inference when computational resources are constrained, without retraining the model.

In biomedical segmentation tasks, U-Net++ has consistently demonstrated improvements over the standard U-Net baseline. Zhou et al. [36] evaluated the architecture across multiple medical imaging datasets including polyp segmentation, cell nuclei detection, and liver lesion segmentation, reporting consistent IoU and Dice improvements over U-Net and other encoder-decoder variants. These gains were most pronounced for small and irregularly shaped structures, a finding of direct relevance to SC segmentation where the target layer is geometrically thin and exhibits natural variability in thickness and boundary regularity across the image.

In the context of OCT segmentation specifically, U-Net++'s dense feature aggregation offers a meaningful architectural advantage over both U-Net and DeepLabV3+. Unlike U-Net,

which relies on single-scale skip connections, U-Net++ progressively refines features across multiple decoder depths before producing the final boundary prediction. Unlike DeepLabV3+, which sacrifices spatial skip connections in favour of global contextual reasoning through ASPP, U-Net++ retains full spatial detail throughout the decoder while simultaneously enriching it through dense feature aggregation. This combination of spatial precision and progressive feature refinement makes U-Net++ a well-motivated architecture for the challenging task of SC boundary delineation in low-contrast, speckle-affected OCT images, and motivates its inclusion as the third architecture evaluated in this project.

5.9 Comparative Analysis of Architecture (U-Net vs DeepLabV3+ vs U-Net++)

The three models compared and evaluated in this project represent distinct design philosophies that produce meaningful trade-offs across different important challenges of SC segmentation. A structured comparison across the dimensions most relevant to this task includes boundary recovery, contextual reasoning, data efficiency, computational cost, and medical imaging validation revealing both why each model is a legitimate candidate and where each is most likely to encounter difficulty.

U-Net’s main strength lies in its explicit boundary recovery mechanism. The symmetric skip connections that define its design and architecture allow full-resolution spatial detail from the encoder to be directly injected into the decoder at each scale, meaning that fine structural information suppressed by downsampling is actively restored rather than approximated [32]. This makes U-Net the most principled choice for thin-layer segmentation tasks, a property empirically confirmed by Roy et al. [33], who demonstrated that skip connection removal caused significant performance drop specifically for the final retinal layer boundaries, a scenario directly analogous to the SC. U-Net’s data efficiency, demonstrated with as few as thirty training images by Ronneberger et al. [32], is also a practical advantage given the limited labelled dataset available in this project. Its main limitation is single-scale processing within each encoder stage, which constrains its ability to simultaneously reason about the SC's local boundary characteristics and its positional relationship to surrounding epidermal layers.

DeepLabV3+ addresses this limitation directly through ASPP, which provides explicit multi-scale feature aggregation in a single module, allowing the network to simultaneously capture fine boundary signals and broad contextual structure [38]. This is likely to be advantageous in regions of the OCT image where the SC's position must be inferred from its relationship to adjacent layers rather than from its own local intensity profile. However, the absence of a strong symmetric decoder means that boundary localisation at full resolution is less precise than in U-Net, as fine spatial detail is not explicitly recovered through skip connections but instead approximated by bilinear upsampling [38]. DeepLabV3+ is also the most computationally demanding of the three architectures and the most dependent on a powerful pre-trained backbone, which has implications for training stability when data is scarce.

U-Net++ occupies a compelling middle ground between these two architectures. Its dense nested skip connections directly address the “semantic gap” limitation of standard U-Net, allowing progressive feature refinement across multiple decoder depths before the final boundary prediction is produced [36]. Unlike DeepLabV3+, which sacrifices spatial skip connections in favour of global contextual reasoning, U-Net++ retains full spatial detail throughout the decoder while simultaneously enriching feature representations through dense aggregation. This combination of spatial precision and progressive refinement makes U-Net++ the most architecturally complete of the three models for thin-layer segmentation, at the cost of higher computational demand relative to U-Net.

The table below summarises these trade-offs:

Dimension	U-Net	DeepLabV3+	U-Net++
Boundary Recovery	Strong, explicit symmetric skip connections	Weak, bilinear upsampling only	Strongest; dense nested progressive refinement
Contextual Reasoning	Limited; single-scale per encoder stage	Strong; explicit ASPP module	Moderate; aggregated through dense connections

Data Efficiency	High; proven with small datasets	Moderate; dependent on deep pre-trained backbone	High; inherits U-Net data efficiency
Computational cost	Low	Moderate	Moderate–High
Medical Validation	Extensive	Moderate	Growing
Primary Risk	Boundary confusion at low contrast regions	Imprecise boundary localisation at full resolution	Higher inference latency

Table 3: Comparative Summary of U-Net, DeepLabV3+, and U-Net++ for Stratum Corneum Segmentation

5.10 Gap and Motivation

Despite the substantial progress made in deep-learning-based OCT segmentation, the application of these methods to skin and to the SC specifically remains an incompletely solved problem. The overwhelming majority of deep learning architectures for OCT segmentation have been developed and benchmarked on retinal imaging datasets, where large, standardised, publicly available annotated collections such as the Duke DME dataset have driven rapid architectural progress [29]. No equivalent resource exists for skin OCT, and the absence of a standardised benchmark means that published results on skin OCT segmentation are difficult to compare directly, as they are evaluated on different datasets, imaging systems, anatomical sites, and annotation protocols [6]. This disparity not only limits the generalisability of reported findings but also means that architectural choices validated on retinal data cannot be assumed to transfer reliably to the structural and optical characteristics of skin.

A more fundamental limitation is that the SC presents a substantially harder segmentation target than the retinal layers on which most deep learning OCT models have been evaluated. Retinal layers are relatively consistent in thickness and optical properties across subjects and are clearly visible at standard OCT resolution. The SC, by contrast, is extremely thin, highly variable across anatomical sites and individuals, and only reliably visible at

ultra-high-resolution—conditions that challenge even well-established architectures. Furthermore, the absence of large publicly available labelled skin OCT datasets means that models must be trained on small, locally annotated collections, amplifying the risk of overfitting and limiting the statistical power of performance evaluations.

This project directly addresses this gap by implementing and systematically comparing three architecturally distinct deep learning models—U-Net, DeepLabV3+, and U-Net++—for SC segmentation in ultra-high-resolution OCT images under a controlled experimental framework designed to isolate the contribution of architectural design choices. By doing so, it contributes both practical segmentation tools and architectural insights that can inform future work in this under-explored but clinically important domain.

6 Methodology & System Design

6.1 Labeling the Dataset

Accurate and consistent annotation of OCT images is a critical prerequisite for supervised deep-learning-based segmentation. In this project, a dataset of 212 OCT B-scan images were manually labelled to identify key skin layer boundaries, with particular focus on the SC.

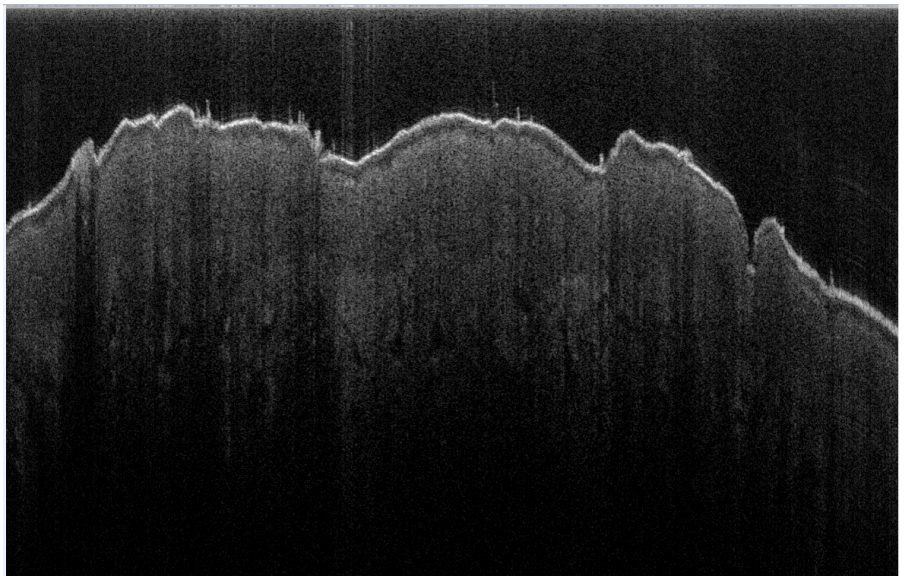


Figure 2: Example of raw OCT image (before labelling)

Figure 2 illustrates an example of a raw OCT image prior to annotation. As observed, the image contains significant speckle noise, intensity variations, and weak boundary contrast between adjacent layers, particularly between the SC and the underlying viable epidermis.

These challenges make fully automated segmentation difficult and highlight the importance of high-quality-ground truth labels.

To generate these labels, a custom MATLAB based annotation tool (OCTlabelTool) was used. This tool enables semi-automated labeling by combining manual user input with edge-based refinement techniques. Initially, OCT image data was loaded into the tool, and a rough estimate of the skin was optionally computed using gradient-based methods to assist the annotation process.

The labeling procedure involved manually placing points along three key anatomical boundaries:

- The outer skin surface (SS)
- The lower boundary of the stratum corneum (SC)
- The dermo epidermal junction (DEJ)

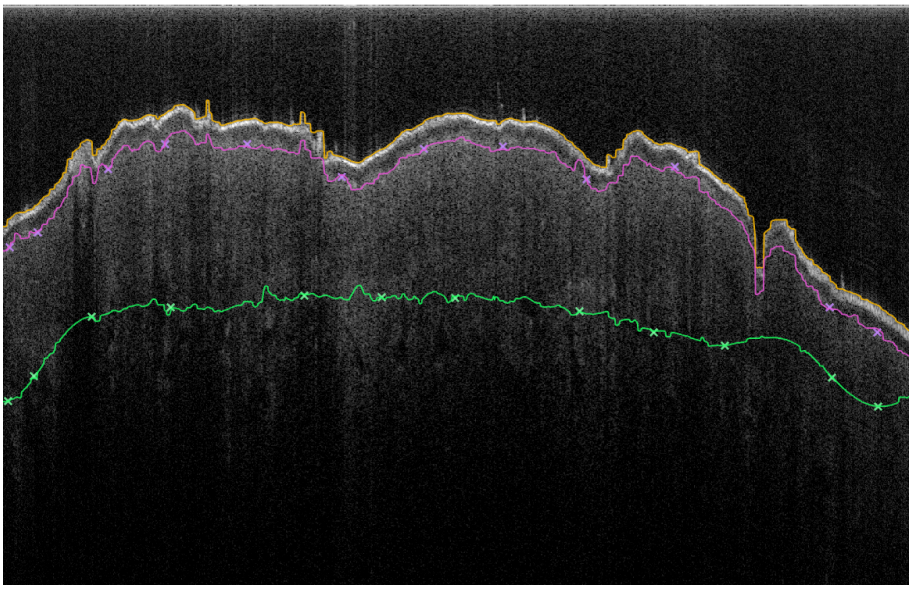


Figure 3: Example of OCT image with labelled skin layers

These points were placed interactively across each image, providing sparse but informative guidance to the tool. Following this, the “Generate Layers & Labels” function was used to automatically refine the boundaries. The tool aligns user-defined points to nearby intensity gradients, performs vertical edge searches in undefined regions, and applies interpolation to generate continuous layer boundaries across the full image width. Additional smoothing is applied to remove unrealistic discontinuities.

Figure 3 shows an example of a labelled OCT image, where the detected layer boundaries are overlaid on the original scan. The skin surface, SC boundary, and DEJ are clearly delineated, providing structured information suitable for training segmentation models.

To ensure annotation quality, a sequential labelling strategy was adopted. The skin surface was first verified and corrected, followed by the SC boundary, and finally the DEJ. This approach reduces error propagation, as deeper layers are dependent on the accuracy of superficial boundaries. All labelled images were visually inspected, and manual corrections were applied where necessary to ensure anatomical consistency across the dataset.

The final output of the labelling process consisted of pixel-wise label maps and layer boundary coordinates, stored in MATLAB structures. Each labelled image was paired with its corresponding annotation mask using filename-based matching to ensure correct supervision and prevent label misalignment. These 212 image-mask pairs were subsequently used as the base dataset for training and evaluating the deep learning segmentation models described in the following sections.

6.2 Performance and Computational Metrics

The metrics used in this project fall into two categories. The first group—comprsing mean pixel accuracy, Intersection over Union (IoU), Dice coefficient, sensitivity, specificity, and precision—are classification-based metrics that evaluate segmentation quality at the pixel level by comparing predicted and ground truth label maps [39].

The second group—parameter count, GFLOPs (Giga Floating Point Operations), and inference latency—are computational efficiency metrics that evaluate the practical deployment requirements of each architecture, providing insight into the trade-off between accuracy and computational cost. Together, these metrics ensure a comprehensive assessment: the classification-based group quantifies the model's ability to delineate the SC and other epidermal layers with high fidelity, while the computational group assesses the real-world feasibility of integrating such models into a clinical OCT imaging pipeline [39].

6.2.1 Pixel Accuracy

Pixel accuracy is the most fundamental metric for evaluating segmentation performance and serves as an intuitive starting point for model comparison. It measures the proportion of

correctly classified pixels across the entire image, providing a straightforward global measure of classification performance. For a binary segmentation problem, pixel accuracy is defined as [39]:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Equation 7: Binary Accuracy

for multi-class segmentation, this can be defined as:

$$Accuracy = \frac{\sum_{i=1}^{K} TP_i}{\sum_{i=1}^{K} (TP_i + FP_i + FN_i)}$$

Equation 8: Multi-class Accuracy

where *K* represents the number of classes, and TP, FP, TN, and FN represent true positives, false positives, true negatives and false negatives respectively.

Pixel accuracy has well-documented critical limitations in the context of medical image segmentation: it is highly susceptible to the effects of class imbalance [39]. In OCT skin images, the background region dominates the pixel distribution, occupying the vast majority of each image, while clinically important features and structures such as the SC, occupy only a small number of pixels vertically. This distributional asymmetry means that a degenerate model predicting background for every pixel could theoretically achieve pixel accuracy exceeding 90%, despite producing no clinically useful segmentation whatsoever. This failure mode, sometimes referred to as the accuracy paradox, is a present concern in thin-layer biomedical segmentation tasks and underscores why pixel accuracy cannot serve as the sole evaluation criterion [40, 42].

In this project pixel accuracy is nonetheless retained as a reported metric because it provides a useful upper-bound reference and enables comparison with prior published work that reports it as a primary measure. Furthermore, when interpreted alongside other metrics such as IoU and Dice, a discrepancy between high pixel accuracy and lower IoU or Dice values is itself informative: it reveals the degree to which a model's apparent performance is inflated by correct background classification rather than accurate detection of target structures. While pixel accuracy provides a useful global reference, it is insufficient as a standalone metric for

thin-layer segmentation due to class imbalance. The following subsection introduces Intersection over Union, which directly addresses this limitation through its overlap-based formulation.

6.2.2 Intersection over Union (IoU)

Intersection over Union, also referred to as the Jaccard Index after its original formulation by Paul Jaccard [40], is one of the most widely adopted and rigorously motivated metrics in semantic segmentation evaluation. It measures spatial overlap between the predicted segmentation mask *P* and the ground truth *G* by computing the ratio of their intersection of their union, thereby directly quantifying how well the model's predicted region coincides with the true anatomical boundary. For a single class, IoU is defined as:

$$IoU = \frac{|P \cap G|}{|P \cup G|} = \frac{TP}{TP + FP + FN}$$

Equation 9: IoU Single Class

For multi-class segmentation, the mean Intersection over Union (mIoU) is computed by averaging the per-class IoU scores across all K classes:

$$mIoU = \frac{1}{K} \frac{\sum_{i=1}^{K} TP_i}{\sum_{i=1}^{K} (TP_i + FP_i + FN_i)}$$

Equation 10: Mean IoU

An important property of IoU that makes it particularly well-suited to thin-layer segmentation is its symmetric penalisation of both over-segmentation and under-segmentation. Mathematically, any false-positive prediction increases the denominator of the IoU expression without contributing to the numerator, and any false-negative similarly reduces the numerator while the denominator remains unchanged. This means that a model which consistently predicts a slightly wider or narrower SC than the ground-truth will be penalised in direct proportion to the magnitude of its boundary errors, regardless of their direction [40]. This symmetric sensitivity to both error types is a significant advantage over pixel accuracy, which penalises false-positives and false-negatives only indirectly through their contribution to the total misclassified pixel count.

In the context of this project, IoU is the main comparative metric for several reasons. First, the SC and skinsurface layers occupy only a small fraction of the total image pixels, meaning that accurate localisation of these thin structures is not adequately rewarded by pixel-level accuracy alone. Second, IoU values are directly interpretable in spatial terms. Third, mIoU is the standard primary metric used in the semantic segmentation benchmarking literature [27, 40], enabling meaningful comparison with published results on related tasks. Fourth, per-class IoU decomposition reveals which specific layer boundaries are most challenging for each model, providing diagnostic insight that aggregate metrics obscure.

An important limitation of IoU is its sensitivity to small absolute errors in thin structures. For a very thin class occupying only a few pixels vertically, even a one pixel boundary displacement can produce a disproportionately large reduction in IoU relative to the same displacement in thicker class, because the denominator of the ratio is small [39]. This scale-sensitivity means that IoU scores for the SC class should be interpreted with awareness that they may appear lower than equivalent quality predictions for thicker classes such as the DEJ, not because the boundary localisation is worse in absolute terms, but because the target region is geometrically smaller. This consideration motivates the joint reporting of IoU alongside the Dice coefficient, which, despite measuring related quantities, distributes this sensitivity differently across its numerator and denominator. While IoU provides a rigorous measure of spatial overlap, its sensitivity to small absolute errors in thin structures motivates the joint reporting of the Dice coefficient, which distributes error sensitivity differently and is more directly connected to the loss function used during model training.

6.2.3 Dice Coefficient

The Dice Similarity Coefficient (DSC), originally proposed by Dice [43] and independently by Sørensen [41] in the context of ecological similarity measurement, has become the dominant overlap metric in biomedical image segmentation evaluation. It measures the harmonic of precision and recall at the pixel-level, and is formally defined as:

$$Dice = \frac{2|P \cap G|}{|P| + |G|} = \frac{2TP}{2TP + FP + FN}$$

Equation 11: Dice

The factor of two in the numerator ensures that perfect overlap between *P* and *G* yields a Dice score of 1.0, while complete disagreement yields 0.0. For multi-class segmentation, the

mean Dice score is computed by averaging per-class values across all *K* classes. The algebraic relationship between Dice and IoU is fixed and monotonic:

$$Dice = \frac{2 \cdot IoU}{1 + IoU}$$

Equation 12: Dice-IoU Relationship

This relationship confirms that Dice and IoU are not independent measures—they are mathematically equivalent in terms of the information they convey about spatial overlap—but they weigh errors differently. Specifically, Dice places relatively more weight on true-positive intersection compared to IoU, making it somewhat less sensitive to large imbalance between foreground and background class sizes [41]. This distributional property makes Dice particularly well-suited to medical image segmentation tasks where target structures are small relative to the image, because it does not penalise small foreground regions as harshly as IoU for the equivalent absolute boundary errors.

The use of Dice as a training loss function, in addition to its role as an evaluation metric, is a particularly important consideration in this project. As discussed in the mathematical foundations section, the Dice loss is defined as:

$$L_{Dice} = 1 - \frac{2\sum \hat{y} \cdot y}{\sum \hat{y} + \sum y}$$

Equation 13: Dice Loss

Where $\hat{y}$ and y denote the predicted probability and ground truth label, summed over all pixels. This formulation directly optimises the spatial overlap between prediction and ground-truth during training, making learning objective directly aligned with the evaluation metric [23]. This alignment between loss function and evaluation metric is considered best practice in thin-layer segmentation because it avoids the situation where a model trained on cross entropy loss achieves low cross-entropy values but poor spatial overlap, particularly for minority classes such as the stratum corneum [23]. In this project, a combined Dice and cross entropy loss was employed during training of all three models to balance sensitivity to thin-layer overlap with stable gradient behaviour during early training epochs.

Clinically, Dice scores are widely used as thresholds for testing the acceptability of automated segmentation for clinical deployment. Values above 0.90 are considered to indicate near-expert agreement, and values between 0.70 and 0.90 indicate clinically acceptable performance, and values below 0.70 suggest performance insufficient for unsupervised clinical use [34, 44]. While Dice captures the aggregate balance between precision and sensitivity, reporting these two components individually provides deeper diagnostic insight into the direction of segmentation errors, as discussed in the following subsections.

6.2.4 Sensitivity (Recall)

Sensitivity, also referred to as recall or the true-positive rate, measures the proportion of all actual positive pixels that are correctly identified as positive by the model [39]. It is formally defined as:

$$Sensitivity = \frac{TP}{TP + FN}$$

Equation 14: Sensitivity

A sensitivity value of 1.0 indicates that every positive pixel in the ground-truth has been detected, while a value approaching 0.0 shows systematic failure to detect the target structure. Sensitivity is therefore a direct measure of the model's detection completeness—it captures the extent to which the segmentation avoids false-negatives, irrespective of how many false-positives are simultaneously generated.

In the context of SC segmentation, sensitivity carries a specific clinical importance. False-negative errors—pixels belonging to the SC that the model fails to identify or find—correspond directly to the regions of the target layer that are missed entirely in the predicted segmentation mask. In the field of dermatological testing, this has major implications: missed regions of the SC would lead to systemic underestimation of layer thickness measurements derived from the segmentation, therefore possibly masking clinically important thinning of the barrier that may be indicative of conditions such as atopic dermatitis, psoriasis, or ichthyosis [5,6]. High sensitivity is therefore not merely a desirable statistical metric but a clinically necessary requirement for the segmentation outputs to be reliable for downstream quantitative analysis.

It is important, however, to interpret sensitivity with specificity and precision rather than in isolation. A model that predicts positive labels for every pixel would achieve a sensitivity of 1.0 trivially, but would be clinically useless due to the extreme false-positive rate. This interdependence between sensitivity and specificity forms a theoretical basis of Receiver Operating Characteristic (ROC) analysis [39], and motivates the joint reporting of both metrics throughout this evaluation.

In practise the relative weighting of sensitivity and specificity errors depends on the clinical application: in screening contexts where missing a positive finding is more costly than a false alarm, sensitivity is prioritised; in confirmatory contexts where false-positives carry significant drawbacks, precision becomes more important. For SC thickness estimation, where both missed detection and over-detection propagate directly into quantitative measurements, a balanced optimisation of both sensitivity and precision is the most ideal objective, as captured by the dice coefficient. Sensitivity alone cannot characterise segmentation quality without consideration of its complementary metric, specificity, which quantifies the model's ability to correctly exclude non-target regions.

6.2.5 Specificity

Specificity, also known as the true-negative rate or selectivity, measures the proportion of all negative pixels that are correctly classified as negative by the models [39]. It is defined as:

$$Specificity = \frac{TN}{TN + FP}$$

Equation 15: Specificity

A specificity of 1.0 shows that no negative pixels are incorrectly labelled as positive—that is, the model generates no false-positives predictions—while lower values show increasing rate of over-segmentation. Specificity is the complementary counterpart to sensitivity: where sensitivity measures the model's ability to find the target structure wherever it is present, specificity measures the model's ability to correctly exclude non-target regions from the positive prediction.

In OCT skin segmentation, achieving high specificity is usually straightforward for most models due to the large size of the background class. Because background pixels constitute the majority of each image, even a model with limited discriminative capability will classify

most of them correctly, producing a high specificity score almost by default. This “ceiling effect” means that specificity differences between architecturally distinct models are typically small in absolute terms, and high specificity alone is not a meaningful discriminator of model performance [39].

Despite this limitation, specificity remains an important metric of the evaluation framework for two reasons. First, the joint interpretation of sensitivity and specificity provides a complete characterisation of the confusion matrix that neither metric can provide alone. A model achieving high sensitivity but low specificity is over-segmenting—detecting the target layer where it does not exist—while a model with high specificity but low sensitivity is under-segmenting—missing portions of the target layer. For the SC, both error types have distinct clinical consequences: over-segmentation inflates apparent layer thickness, while under-segmentation deflates it. Only by examining both metrics simultaneously can the direction and nature of segmentation errors be properly characterised [39].

Second, in the per-class analysis, the specificity of thin-layer classes such as the SS and SC is more informative than that of the background, because the relevant negative pixels for these classes include all non-SC tissue, where false-positive predictions represent anatomically implausible boundary locations that a robust model should confidently exclude. Together, sensitivity and specificity characterise the model's behaviour from the perspective of the ground-truth. Precision complements this by characterising model behaviour from the perspective of the predictions themselves, as discussed in the following subsection.

6.2.6 Precision

Precision, also referred to as the positive predictive value (PPV), measures the proportion of pixels predicted as positive by the model that are truly positive in the ground-truth [39]. It is formally defined as:

$$Precision = \frac{TP}{TP + FP}$$

Equation 16: Precision

A precision of 1.0 indicates that every pixel predicted as belonging to the target class is correct—there are no false-positives—while lower values indicate that an increasing proportion of positive predictions are spurious. Precision therefore captures the reliability or

purity of the model's positive predictions, independently of how many true-positive pixels were missed.

In the context of SC segmentation, precision measures the degree to which the model's predicted boundary region corresponds to the true anatomical boundary rather than including surrounding tissue incorrectly. Low precision corresponds to over-segmentation—the model predicts SC pixels in regions belonging to the viable epidermis or background—which would inflate apparent SC thickness in quantitative analyses. This error type is particularly problematic in regions of low inter-layer contrast, where the boundary between the SC and the underlying viable epidermis is ambiguous due to speckle noise or similar optical properties of adjacent layers [6,8]. In such regions, a model with insufficient discriminative capacity may expand its predicted boundary into adjacent tissue to maximise sensitivity, at the cost of reduced precision.

The relationship between precision and sensitivity, and their aggregation into the Dice coefficient, can be expressed formally as:

$$Dice = \frac{2 \times Precision \times Sensitivity}{Precision + Sensitivity}$$

Equation 17: Dice as Harmonic Mean

This formulation reveals that the Dice coefficient is the harmonic mean of precision and sensitivity. The harmonic mean is more sensitive to low values than the arithmetic mean, meaning that a model achieving very high sensitivity but very low precision—or vice versa—will receive a substantially penalised Dice score relative to a model that achieves balanced performance on both [43]. This property makes the Dice coefficient a natural aggregate of precision and sensitivity for evaluating boundary quality, but it also means that identical Dice scores can arise from fundamentally different combinations of precision and sensitivity. Reporting precision and sensitivity individually therefore provides important diagnostic information that Dice alone does not convey: it allows direct identification of whether each model tends toward under-detection or over-detection of the SC, which has direct implications for the direction of any systematic bias in thickness measurements derived from the segmentation outputs.

In this project, per-class precision values are reported alongside sensitivity and Dice to enable this decomposed analysis. The comparison across U-Net, DeepLabV3+, and U-Net++ reveals whether architecturally distinct boundary recovery mechanisms—skip connections in U-Net, multi-scale ASPP in DeepLabV3+, and dense skip connections and nested encoder-decoder structure in U-Net++—produce different characteristic error patterns in terms of over- versus under-segmentation of thin epidermal layers.

The classification-based metrics discussed in sections 6.2.1 through 6.2.6 collectively characterise pixel-level segmentation quality. To evaluate the practical feasibility of these models in a clinical setting, the following sections introduce the computational efficiency metrics used to assess hardware requirements and inference speeds.

6.2.7 Parameters

The number of parameters in a deep learning model refers to the total number of learnable weights and biases across all layers of the network. These parameters are updated during training through backpropagation and represent the model's learned knowledge about the task. A higher parameter count generally shows a more expressive model capable of learning complex feature representations, but also increases memory requirements and the risk of overfitting, mainly when training data is limited, In the context of this project, reporting the parameter count for each model provides insight into model complexity and helps contextualise performance differences. A model achieving higher accuracy with fewer parameters is considered more efficient and better suited to resource constrained deployment situations [26] such as clinical workstations or embedded medical devices.

6.2.8 GFLOPs

GFLOPs, or Giga Floating Point Operations, measure the computational cost of a single forward pass through the network, expressed in billions of floating point operations. Unlike parameter count, which measures model size, GFLOPs directly quantify the amount of computation required to process one input image. This is an important practical consideration in medical imaging applications where inference speed and hardware efficiency matter. A model with high GFLOPs requires more computational resources per prediction, which can limit its applicability in real-time or high-throughput clinical settings. Comparing GFLOPs

across architectures therefore provides a meaningful measure of computational efficiency that complements the accuracy-based metrics reported in this project.

6.2.9 Latency

Latency refers to the time taken by the model to process a single input image and produce a segmentation output, measured in milliseconds (ms). While GFLOPs capture the theoretical computational cost, latency reflects the real-world inference speed under actual hardware conditions. Lower latency is particularly important in clinical applications where rapid feedback is required, such as real-time OCT-guided procedures or high-throughput screening of large image datasets. In this project, latency was measured by averaging inference time over 100 forward passes on a CPU, providing a hardware-agnostic comparison that reflects the intrinsic speed characteristics of each architecture rather than GPU-specific optimisations.

6.3 Data Preprocessing and Training Configuration

6.3.1 Unified Dataset Preparation and Preprocessing

A well-designed preprocessing pipeline is essential for accurate and successful results. A consistent preprocessing pipeline was applied across all segmentation models, including U-Net, DeepLabV3+, and U-Net++. This ensures that the performance differences between models are traced down to the architectural variations rather than inconsistencies in data handling.

As mentioned in Section 6.1, the dataset consists of 212 manually labelled OCT images paired with corresponding pixel-level annotation masks representing four classes: Background, Skin Surface (SS), Stratum Corneum (SC), and the Dermo-Epidermal Junction (DEJ). Image and mask pairing was validated using filename-based matching to ensure correct supervision and prevent label misalignment.

All the images were resized to a fixed resolution of 512×512 pixels to maintain consistency across models and enable batch processing. Masks were resized using nearest-neighbour interpolation to preserve discrete class labels without introducing interpolation artefacts.

To improve image quality and enhance the visibility of low-contrast structures, Contrast Limited Adaptive Histogram Equalisation (CLAHE) was applied during preprocessing. This step is particularly important for OCT imaging, where tissue boundaries often show weak intensity gradients and are affected by speckle noise.

Following contrast enhancement, all images were normalised using ImageNet mean and standard deviation values to match the distribution expected by pre-trained encoder backbones. This allows effective transfer learning from models trained on large-scale natural image datasets.

6.3.2 Data Augmentation Strategy

To improve model generalisation and reduce overfitting, data augmentation was applied during training. The following transformations were applied across all models:

- Horizontal flipping
- Rotation ($\pm$ 10° degrees)
- Shift, scale, and rotate transformations
- Brightness and contrast variation

For DeepLabV3+, additional stochastic augmentations were applied, including elastic deformation and Gaussian noise injection, to improve robustness to structural variability and imaging artefacts.

Following augmentation, the effective dataset size was expanded from 212 to 1,272 image-mask pairs before being divided into training, validation, and test splits, providing a more diverse and representative dataset to improve model generalisation.

6.3.3 Dataset Splitting and Loading Strategy

Following the augmentation described in section 6.3.2, the 1,272 image-mask pairs were divided into training, validation, and test sets using a 70-15-15 split, resulting in 890 training images, 191 validation images, and 191 test images. This ensures enough data for model learning while maintaining independent subsets for hyperparameter tuning and final evaluation.

PyTorch DataLoader was used to handle batch processing efficiently, with shuffling applied to the training set to reduce learning bias. Parallel data loading was enabled to improve computational efficiency during training.

6.3.4 Training Configuration and Experimental Control

The same training configuration was used across all models to ensure a fair and controlled comparison. This includes identical batch size, optimisation strategy, and learning rate schedule across experiments.

Training was performed using the AdamW optimiser with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-3}. A cosine annealing learning rate scheduler was used to gradually reduce the learning rate over time, promoting stable convergence and improved generalisation.

A hybrid loss function combining weighted cross-entropy and Tversky loss was used across all models. This formulation addresses class imbalance by assigning higher importance to underrepresented classes such as SS and SC, while also optimising spatial overlap between predicted and ground truth masks.

Early stopping with a patience of 12 epochs was implemented based on validation performance to prevent overfitting and ensure optimal model selection. Gradient clipping with a threshold of 1.0 was applied to improve training stability.

The training parameters used are summarised below in table 4:

Parameter	Value
Image Size	512 x 512
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-3}

Scheduler	Cosine Annealing
Epochs	40
Batch Size	8
Loss Function	Weighted Cross-Entropy and Tversky Loss
Gradient Clipping	1.0
Early Stopping Patience	12
Train/Val/Test Split	70/15/15

Table 4: Shared Training Hyperparameters Across All Models

6.3.5 Experimental Consistency Across Models

To ensure a fair comparison between U-Net, DeepLabV3+, and U-Net++, all models were trained using the same preprocessing pipeline and optimisation strategy. Any differences in performance can therefore be primarily attributed to architectural design choices, such as skip connection strategy and feature aggregation mechanisms. The only notable controlled variations between models were architectural modifications, encoder backbone depth, and model-specific augmentation strategies — particularly for DeepLabV3+, where stronger augmentation was applied to improve robustness to speckle noise and structural variability.

7 Experimental Results & Discussion

7.1 U-Net Implementation

The U-Net architecture was implemented as the baseline deep learning model for the segmentation of Optical Coherence Tomography (OCT) skin images. While the theoretical encoder–decoder structure was introduced in the literature review, this implementation was adapted to address the specific challenges associated with OCT imaging, including speckle noise, low contrast between adjacent skin layers, and the presence of thin anatomical structures such as the Skin Surface (SS), Stratum Corneum (SC), and the Dermo-Epidermal Junction (DEJ). These characteristics make precise boundary detection difficult and increase the likelihood of misclassification, particularly for thin and low-contrast regions.

To improve feature extraction, a pre-trained encoder was used. Transfer learning from large-scale natural image datasets enables the model to capture general low-level features such as edges, gradients, and textures, which remain relevant in medical imaging contexts. However, due to the domain shift between natural and OCT images, additional considerations were required to ensure accurate localisation. In particular, the segmentation of thin layers required the model to preserve spatial information throughout the network, as these structures are easily lost during downsampling in the encoder.

The U-Net architecture addresses this through skip connections between encoder and decoder layers, which allow high-resolution spatial features to be retained and combined with deeper semantic information. This is particularly important for reconstructing fine anatomical boundaries and improving segmentation accuracy in regions with weak contrast.

The model was trained using the shared preprocessing and training settings described in section 6.3.

7.1.1 Training Performance and Convergence

The training process showed stable convergence across different evaluation metrics. As shown in Figure 4, the U-Net achieved consistent improvement in segmentation performance over epochs, with no evidence of major instability or divergence.

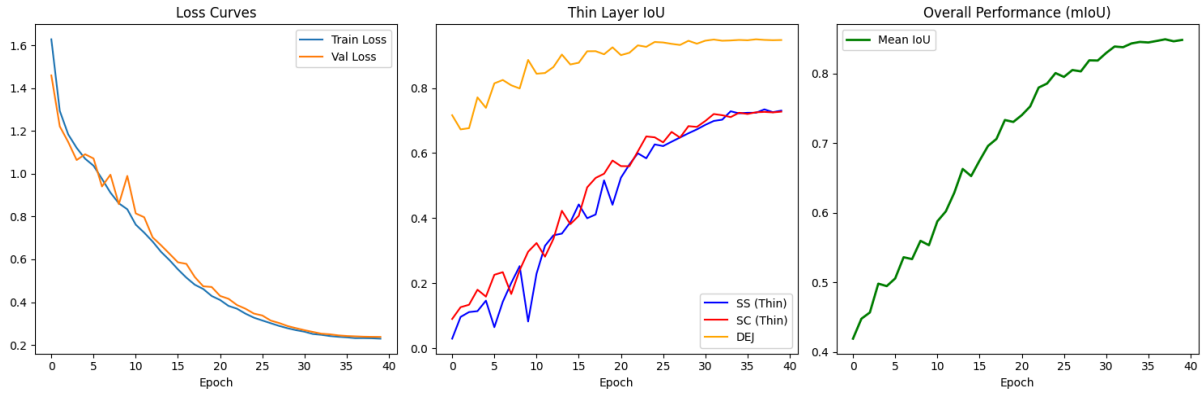


Figure 4: U-Net Training Performance and Convergence

The loss decreased in both the training and validation sets. Additionally, the Intersection over Union (IoU) increased across all classes as the number of epochs increased, indicating that the model successfully learned meaningful feature representations without severe overfitting.

7.1.2 Quantitative Results

The overall quantitative performance of the U-Net model is summarised in Table 4. The model achieved a high mean pixel accuracy of 98.66%, indicating that the majority of pixels were correctly classified. However, pixel accuracy alone is not sufficient for evaluating segmentation performance, particularly in imbalanced datasets where background pixels are the majority.

More informative metrics such as Mean IoU (0.8488) and Dice coefficient (0.9136) show strong overlap between predicted and ground truth segmentations. These values indicate that the model performs well in capturing the general structure of the skin layers.

Sensitivity (0.9537) and specificity (0.9952) results further confirm that the model is effective in identifying true positive regions while maintaining a low positive rate. Precision (0.8797) is slightly lower, suggesting that some oversegmentation occurs, mainly in challenging regions.

Metric	Value
Mean Pixel Accuracy	98.66%
Overall Mean IoU	0.8488
Overall Mean Dice	0.9136
Overall Sensitivity	0.9537
Overall Specificity	0.9952
Overall Precision	0.8797

Table 5: U-Net Overall Metric Table

Overall, these results suggest that U-Net provides strong baseline performance for OCT image segmentation, with good generalisation across the dataset.

7.1.3 Class-Wise Analysis

A more detailed evaluation of the model performance is provided through class-wise metrics in Table 5. These results show how segmentation performance varies across different skin structures.

The background class achieved the highest performance, with an IoU of 0.989 and Dice coefficient of 0.995. This is expected, as background regions are more homogeneous and occupy a large portion of the image, making them easier to classify.

The DEJ class also achieved strong performance, with an IoU of 0.950 and Dice score of 0.974. This suggests that the model is able to capture this boundary effectively, likely due to its relatively distinct structural characteristics compared to surrounding layers.

Comparatively, the SS and the SC classes showed lower performance, with IoU values of 0.741 and 0.738. These layers are thin and have low contrast compared to adjacent regions, making them more difficult to segment accurately. Despite high sensitivity values (above 0.92), the lower precision shows that the model tends to over-predict these classes, leading to false-positives.

Class	IoU	Dice	Sensitivity	Specificity	Precision
Background	0.989	0.995	0.991	0.994	0.998
SS	0.741	0.851	0.921	0.998	0.791
SC	0.738	0.849	0.929	0.997	0.781
DEJ	0.950	0.974	0.981	0.993	0.968

Table 6: U-Net Per-Class Results

This imbalance between sensitivity and precision suggests that while the model is able to detect these layers, it struggles to precisely localise their boundaries. This is a known limitation of standard U-Net models when applied to detailed segmentation tasks.

7.1.4 Qualitative Results

Qualitative evaluation provides further insight into the model's behaviour. As shown in Figure 5, the predicted segmentation maps generally align well with the ground truth labels, especially for larger and more distinct regions such as the background and DEJ.

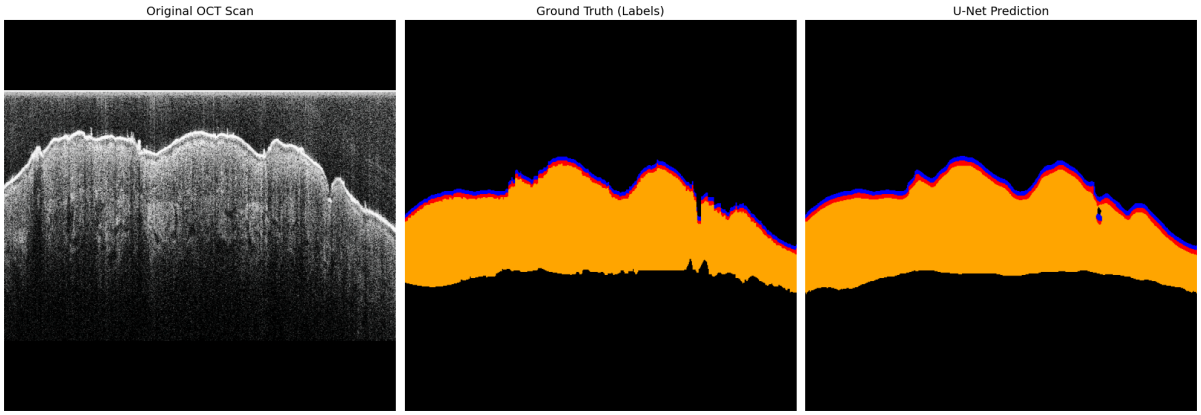


Figure 5: U-Net Ground Truth vs Prediction

However, discrepancies are observed in thinner layers such as SS and SC, where the model predicts blurred or smoothed out shifted boundaries. In some cases shown, these layers appear thicker in the predictions than in ground truth, which is consistent with the over-segmentation shown in the quantitative analysis.

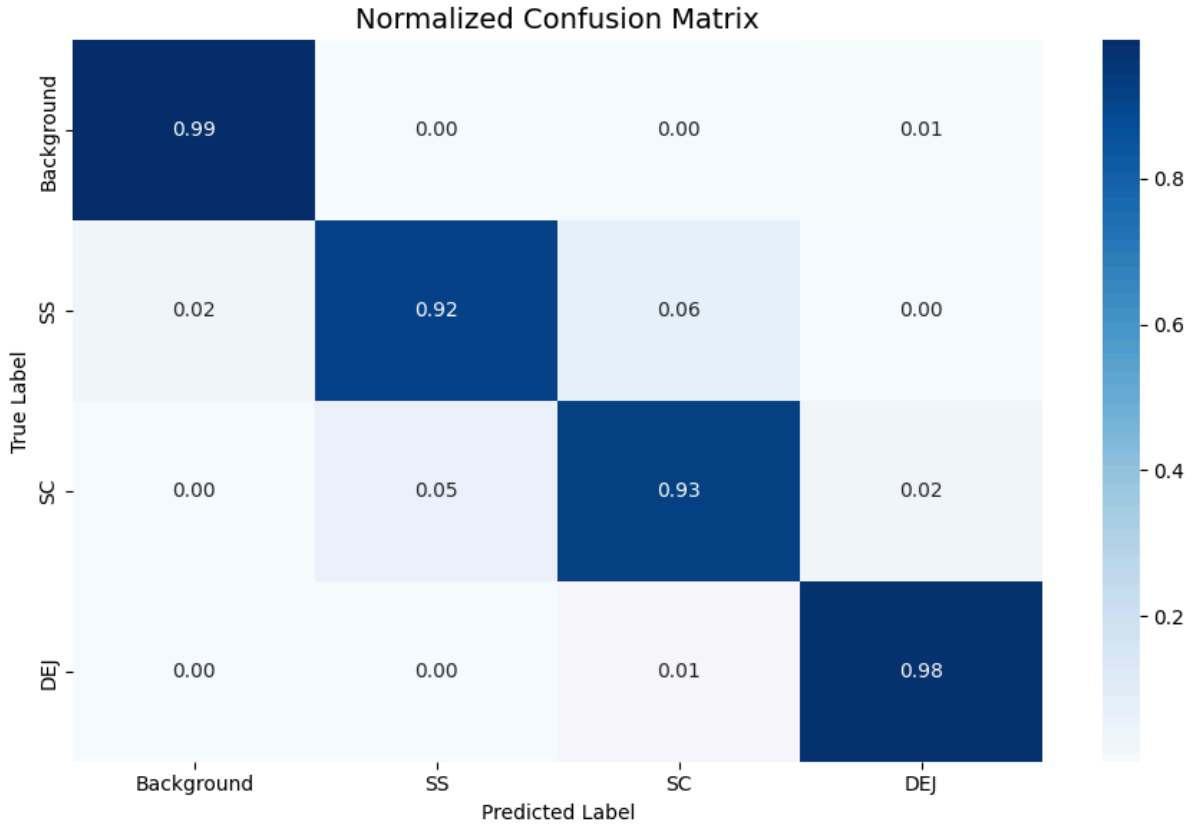


Figure 6: U-Net's Confusion Matrix

The confusion matrix in Figure 6 shows how often the model predicts the correct class versus an incorrect one. The diagonal entries represent correct classifications. The model performs excellently at identifying Background (99%) and DEJ (98%), and also predicts SS (92%) and SC (93%) well, though with slightly lower confidence than the other classes.
The off-diagonal entries represent misclassifications. The most significant confusion is between SS and SC, where the model predicts SC when the true label is SS approximately 6% of the time. This is likely because these two regions are physically adjacent and visually similar in OCT images.

7.2 DeepLabV3+ Implementation

DeepLabV3+ was implemented as a comparative segmentation model to evaluate the impact of multi-scale contextual feature extraction on Optical Coherence Tomography (OCT) skin image segmentation. Unlike the U-Net baseline, this model uses atrous convolution and spatial pyramid pooling to capture contextual information at different spatial scales.

Unlike the other models, DeepLabV3+ used a ResNet-50 encoder pre-trained on ImageNet compared to ResNet-34 for the other models, allowing the model to learn deeper and more complex feature representations. This improves the model's ability to capture global structural information present in OCT images.

As mentioned in the literature review, the Atrous Spatial Pyramid Pooling (ASPP) module enables the extraction of features at different receptive fields, which is useful for identifying broader tissue structures and contextual relationships between layers. However, due to the relatively lightweight decoder and reduced reliance on skip connections, the model places less importance on fine spatial reconstruction.

To address class imbalance, a weighted cross-entropy loss function was applied, with higher weights assigned to underrepresented classes such as SS, SC, and DEJ. In addition, stronger data augmentation strategies were applied during training, including elastic deformation, Gaussian noise injection, and brightness and contrast variation. These were introduced to improve model robustness to speckle noise and structural variability commonly present in OCT images. The model was trained using the shared preprocessing and training configuration described in Section 6.3.

7.2.1 Training Performance and Convergence

Figure 7 shows different convergence curves. The loss curve demonstrates stable convergence across epochs, with a consistent decrease in both training and validation loss. The model also generalised well to unseen data without overfitting, as indicated by the validation loss remaining consistently at or below the training loss throughout training.

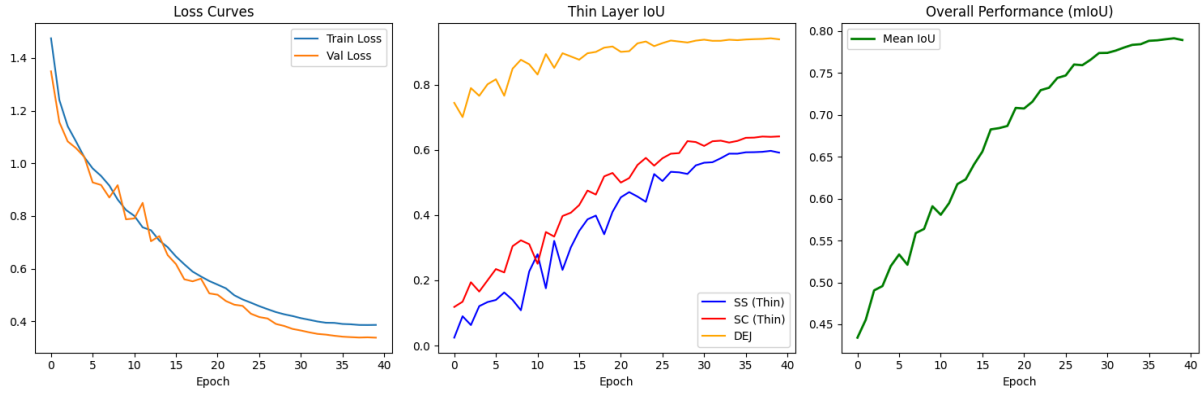


Figure 7: DeepLabV3+ Training Performance and Convergence

The thin layer IoU measures the overlap between the predicted mask and the ground truth. As shown in Figure 7, the DEJ class started strong and reached near-perfect performance (~0.96 IoU) rapidly. Comparatively, the model struggled during the first 10 epochs with the thin layers SS and SC, but steadily improved, eventually reaching 0.58–0.65.

The third curve shows the model's overall mean IoU (mIoU), which is the average IoU across all classes. The curve shows a smooth, steady upward trajectory, ending at approximately 0.79 mIoU.

7.2.2 Quantitative Results

The overall performance of the DeepLabV3+ model is summarised in Table 6. The model achieved a mean pixel accuracy of 98.31%, showing strong overall classification performance across the dataset.

The mean IoU of 0.7929 and Dice coefficient of 0.8740 show good segmentation accuracy, with a high degree of overlap between predicted and ground truth regions. These metrics suggest that the model is effective in capturing the general structure of the skin layers.

Metric	Value
Mean Pixel Accuracy	98.31%
Overall Mean IoU	0.7929
Overall Mean dice	0.8740
Overall Sensitivity	0.9379
Overall Specificity	0.9938
Overall Precision	0.8259

Table 7: DeepLabV3+ Overall Results

Sensitivity (0.9379) and specificity (0.9938) are both high, demonstrating that the model is able to correctly identify true positive regions while maintaining a low rate of false positives. However, the precision value (0.8259) is lower, showing that some oversegmentation occurs, mainly in regions where class boundaries are less distinct.

Overall, these results indicate that the model performs well in segmenting the main anatomical structures present in OCT images, while some inaccuracies remain in regions with complex boundaries.

7.2.3 Class-Wise Analysis

Class-wise performance metrics are presented in Table 7 and provide a more detailed understanding of segmentation performance across different anatomical regions.

The background class achieved high performance, with an IoU (0.986) and Dice Score of (0.993). This is expected due to the relatively large and homogeneous nature of background regions.

The performance on the SS and SC classes is lower, with IoU values of 0.595 and 0.650 respectively. These layers are thin and exhibit low contrast relative to neighbouring regions, making them more difficult to segment accurately. While sensitivity remains high for these classes (0.901 and 0.899), showing that the model successfully captures the majority of true

positive pixels, the lower precision values (0.636 and 0.701) suggest a higher rate of false positives at the boundaries.

Class	IoU	Dice	Sensitivity	Specificity	Precision
Background	0.986	0.993	0.988	0.992	0.998
SS	0.595	0.746	0.901	0.995	0.636
SC	0.650	0.788	0.899	0.995	0.701
DEJ	0.940	0.969	0.970	0.993	0.969

Table 8: DeepLabV3+ Per Class Results

This disparity is a known characteristic of thin structure segmentation; because the total pixel area of the SS and SC layers is small, even a slight spatial displacement or boundary error of one or two pixels largely penalises the IoU and precision scores, while having little impact on sensitivity and specificity.

7.2.4 Qualitative Results

As shown in Figure 8, the predictions appear smoother than the ground truth. The ground truth shows that the layers have jagged, natural edges. This explains why the precision results for SS (0.636) and SC (0.701) are lower.

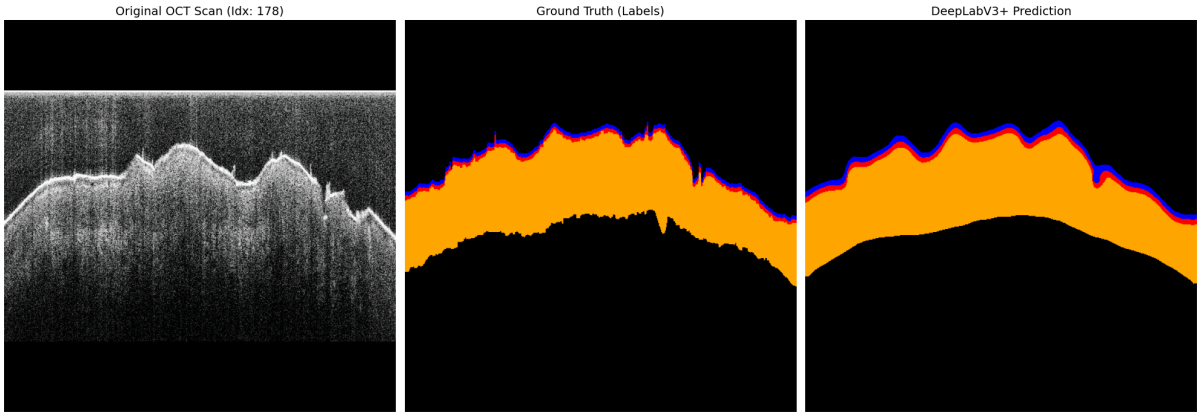


Figure 8: DeepLabV3+ Ground Truth vs Prediction

Because these layers are only a few pixels thick, the model's tendency to produce a smooth boundary rather than following every microscopic variation leads to a high number of false positive and false negative pixels along the boundaries. In a thin structure, these boundary errors represent a large percentage of the total layer area. Therefore, while the model is anatomically correct — meaning it has high sensitivity and places the layers in the correct

stacking order — the IoU is penalised by this lack of edge-matching precision. Moreover, the DeepLabV3+ model uses atrous (dilated) convolutions and a decoder that often favors structural continuity over high frequency detail. This results in the “cleaner” look seen in the prediction, which is visually appealing and likely enough for clinical measurement of layer thickness, which naturally deviates from pixel-perfect irregularity of the manually annotated ground truth.

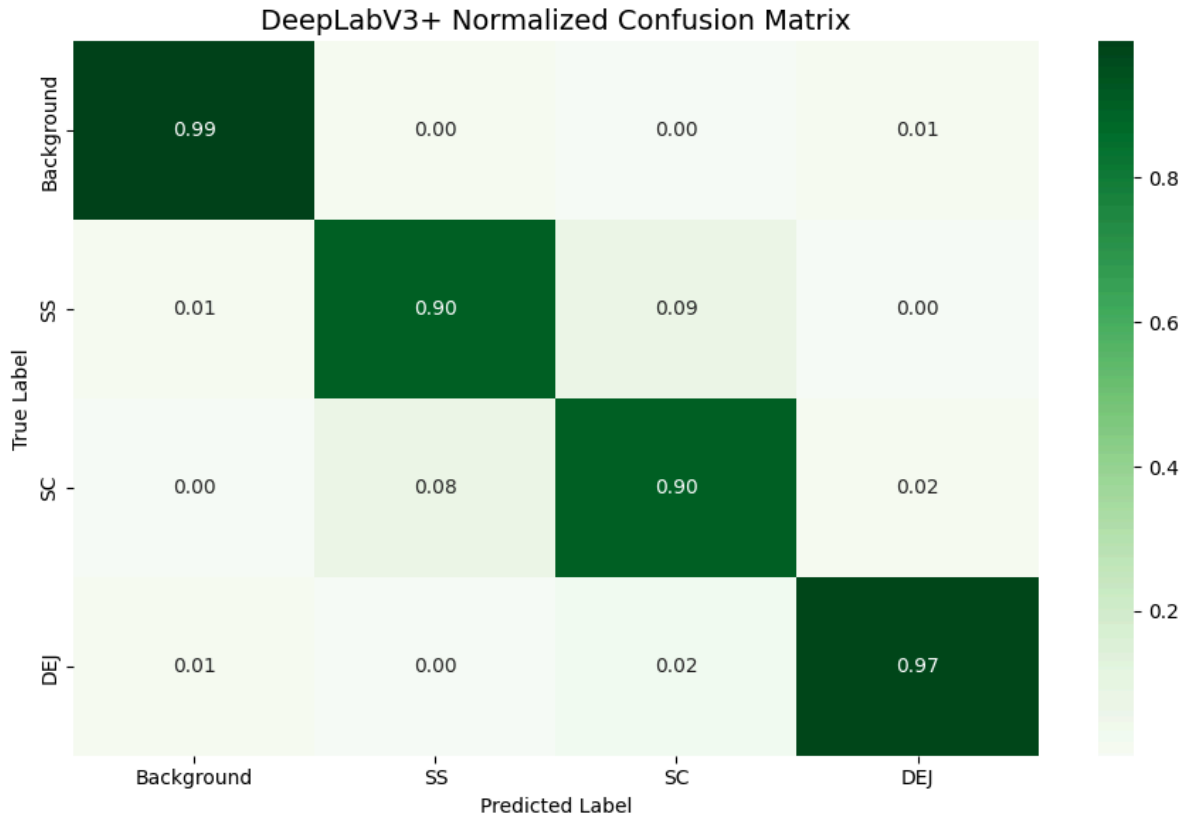


Figure 9: DeepLabV3+ Confusion Matrix

The confusion matrix in Figure 9 confirms the observations from the visual predictions: the model is effective at broad anatomical localisation but faces challenges with thin regions. The confusion matrix shows high categorical accuracy, particularly for the Background (0.99) and DEJ (0.97) classes. The main source of error is confusion between the SS and SC layers, with misclassification rates of 9% and 8% respectively. This localised confusion is due to the adjacency of these thin layers and the low contrast at their interface, which presents a challenge for precise boundary detection despite the model's high overall sensitivity.

7.3 U-Net++ Implementation

The U-Net++ model was implemented to improve segmentation performance through enhanced feature fusion and refined localisation of anatomical boundaries. Unlike the standard U-Net, U-Net++ introduces nested and dense skip connections between the encoder and decoder, which reduce the semantic gap between feature maps at different depths.

These redesigned skip pathways allow the model to progressively refine feature representations before reconstruction, leading to improved segmentation accuracy particularly for structures with complex and thin boundaries.

The model was implemented using the segmentation-models-pytorch framework with a ResNet-34 encoder pre-trained on ImageNet. This provides strong feature extraction while maintaining consistency with the standard U-Net model.

To further enhance feature representation, spatial and channel squeeze-and-excitation attention modules were incorporated within the decoder. These mechanisms enable the network to emphasise informative regions and suppress non-important background features, which is important in OCT images where contrast between layers is limited.

A combined loss function was used to improve segmentation performance on imbalanced classes. Cross-entropy loss with class weighting was applied alongside Tversky loss, allowing the model to better handle thin anatomical structures by penalising false negatives more strongly.

All preprocessing steps, data augmentation methods, and training settings follow the unified setup described in Section 6.3.

7.3.1 Training Performance and Convergence

The performance plots in Figure 10 show stable convergence across all metrics. Both the training loss and validation loss decrease rapidly until approximately epoch 20, after which they taper off into a stable plateau. Furthermore, the validation loss closely tracks the training loss with no significant divergence between them, indicating that the model is generalising well to unseen data and is not overfitting.

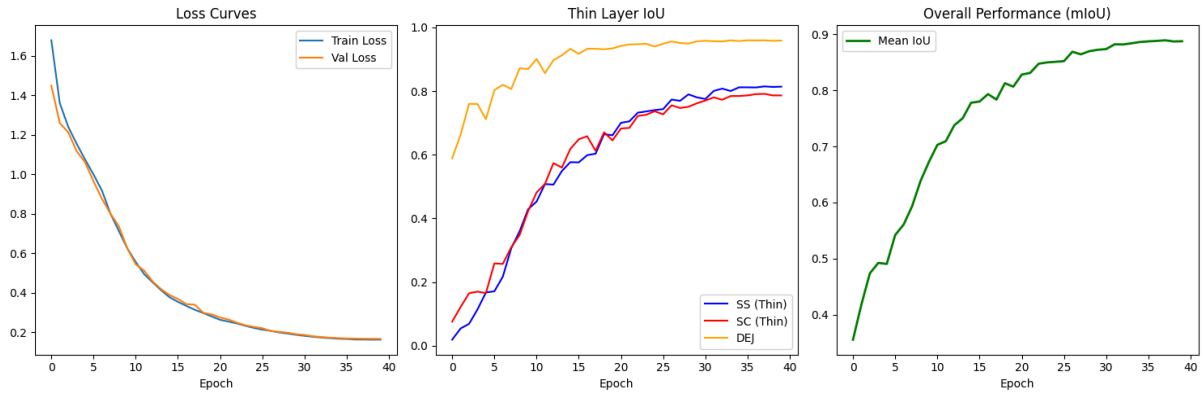


Figure 10: U-Net++ Training Performance and Convergence

The thin layer IoU tracks the IoU for specific structures (SS, SC, and DEJ). At the start, the model struggled with the thin layers SS and SC, but improved steadily, with both layers ending between 0.80 and 0.85, demonstrating that the model handles difficult thin geometries well. The DEJ was the easiest structure for the model to learn, starting at an IoU of approximately 0.6 and reaching near-perfect performance of ~0.95.

The final chart shows the overall mean IoU (mIoU), the primary indicator of the model's segmentation accuracy. The model reaches a peak mIoU of ~0.89. The steepest gains occur within the first 15 epochs, and after epoch 30 the gains become marginal, suggesting that training beyond 40 epochs would likely yield diminishing returns.

7.3.2 Quantitative Results

Table 8 below shows U-Net++'s overall performance across classification metrics.

Metric	Value
Mean Pixel Accuracy	98.99%
Overall Mean IoU	0.8860
Overall Mean dice	0.9372
Overall Sensitivity	0.9670
Overall Specificity	0.9962
Overall Precision	0.9108

Table 9: U-Net++ Overall Results

The mean IoU (0.8860) and Dice coefficient (0.9372) show that the model performed well for the segmentation task. A Dice score of 0.93 indicates a high overlap between the predicted masks and ground truth labels. Furthermore, since Dice is more sensitive to inner region agreement than IoU, this result suggests the model is effective at capturing the core shape of the structures.

The mean pixel accuracy (98.99%) is high, but can be misleading depending on how much background occupies the field of view. In instances where the background dominates, a model can achieve high accuracy by correctly classifying background pixels while performing poorly on the regions of interest. However, when paired with high IoU and Dice scores, it confirms that the model is performing accurately across all classes.

Sensitivity (0.9670) indicates that the model detects 96.7% of actual positive pixels, with a low rate of false negatives, which is important in a medical context. A precision of 0.9108 suggests some tendency towards oversegmentation, meaning the model occasionally labels background pixels as part of a target structure. The specificity of 0.9962 confirms that the model is highly accurate at identifying background pixels with very few false positives.

7.3.3 Class-Wise Analysis

Table 9 presents the per-class breakdown of metrics, providing a more detailed understanding of the model's overall performance.

Class	IoU	Dice	Sensitivity	Specificity	Precision
Background	0.991	0.995	0.992	0.995	0.999
SS	0.805	0.892	0.947	0.998	0.843
SC	0.791	0.883	0.944	0.998	0.829
DEJ	0.957	0.978	0.985	0.994	0.972

Table 10: U-Net++ Per Class Results

The background class achieved an IoU of 0.991, indicating that the model has minimal confusion over non-skin regions. The DEJ achieved a strong IoU of 0.957 for a specific boundary layer. Across all remaining metrics, both classes performed well, indicating that the model's predictions are highly accurate and consistent for well-defined regions. This suggests

that the model reliably captures clear structural features, particularly where there is strong contrast between adjacent layers.

The SS and SC classes show lower performance compared to the DEJ, but remain strong results for thin layers. The sensitivity values (above 0.94) versus precision values (~0.83–0.84) for these two layers indicate that the model is effective at detecting SS and SC but tends to slightly overpredict their thickness, likely due to boundary bleeding between adjacent layers.

7.3.4 Qualitative Results

The figures below confirm that the model is visually accurate. By comparing the U-Net++ predictions to the ground truth OCT scans, it can be observed that the model maintains structural integrity across complex anatomical boundaries. The confusion matrix reinforces these observations, with the majority of predictions lying along the diagonal, indicating a high rate of correct classifications across all classes with minimal confusion between them.

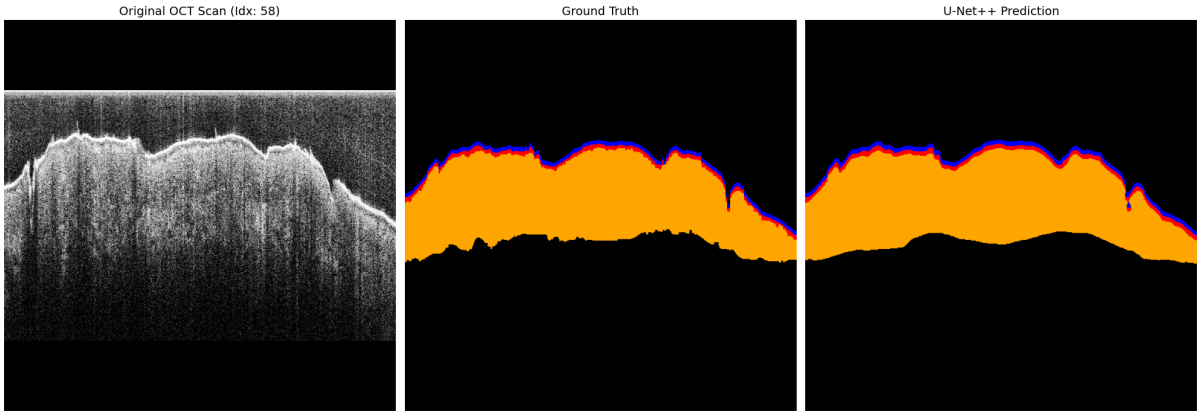


Figure 11: U-Net++ Ground Truth vs Prediction

As shown in Figure 11, the U-Net++ prediction captures the undulating, jagged surface of the skin almost identically to the ground truth, demonstrating excellent morphological capture. Moreover, the DEJ is consistent and does not disappear even in areas where the OCT signal is noisier or darker. The SS and SC are also captured well, although there are some minor discrepancies in the prediction's smoothness and boundary tracking in the regions where the scan dips. This is likely where the model struggled with a sharp drop in signal intensity in the original scan. Overall, the model demonstrates high noise immunity, producing smooth, continuous segmentation masks despite the grainy input.

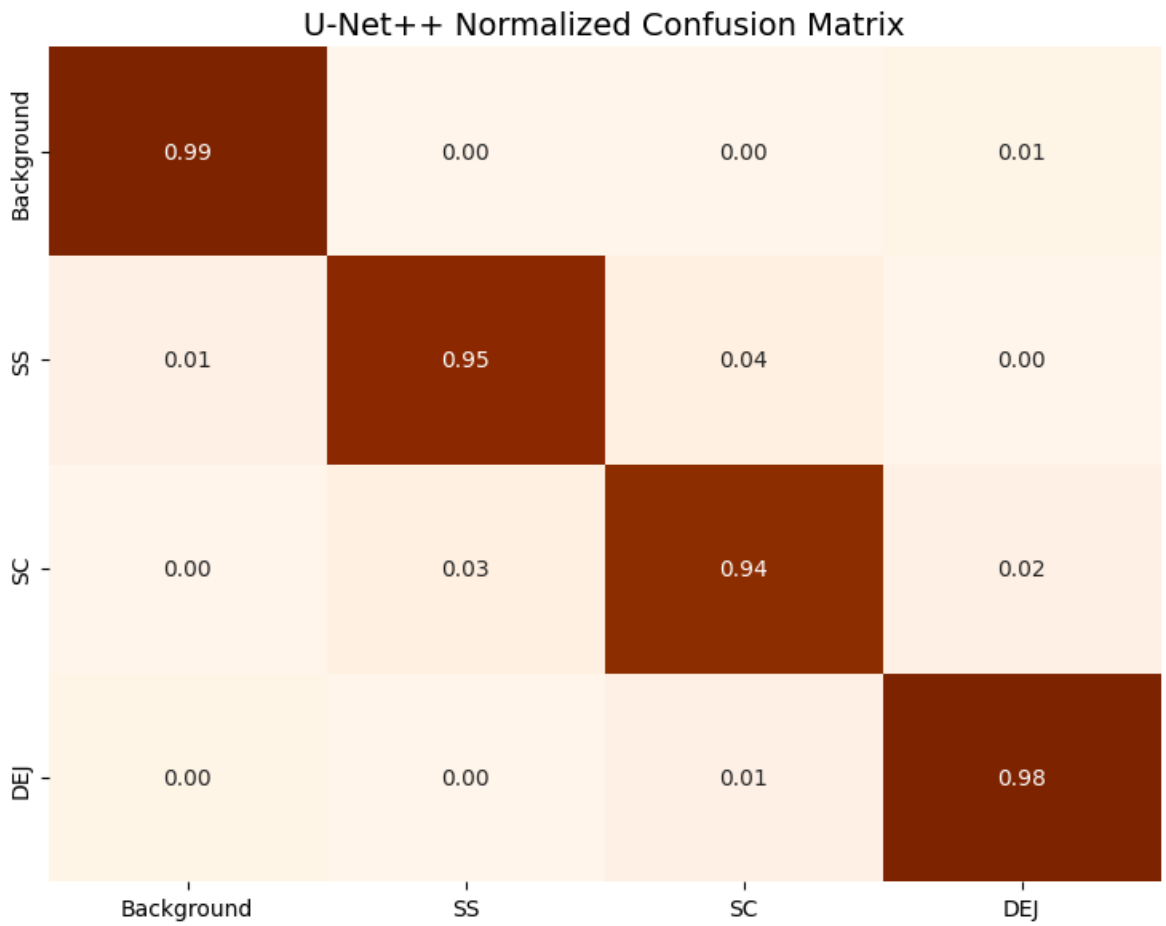


Figure 12: U-Net++ Confusion Matrix

The confusion matrix is row-normalised, with minor discrepancies due to rounding. It shows strong diagonal dominance, with values of 0.99, 0.95, 0.94, and 0.98, indicating that the model correctly identifies the vast majority of pixels for each class.

The SS versus SC confusion confirms the findings discussed in Section 7.3.3, where 4% of actual SS pixels are predicted as SC and 3% of actual SC pixels are predicted as SS. This is attributed to boundary ambiguity between these adjacent layers, where pixels from one class are misclassified as the other due to their similar visual characteristics and close spatial proximity. The DEJ shows almost no confusion with the SS and background classes, which is notable given how thin this interface is.

7.4 Results Comparison Between Models

This section brings together the results from all three models to provide a cross model comparison across segmentation accuracy, SC specific performance, and computational

requirements. The three tables below summarise these comparisons and form the basis of the analysis presented here.

7.4.1 Overall Segmentation Performance

Across all classification metrics, U-Net++ consistently achieved the strongest performance, followed by U-Net, with DeepLabV3+ performing the lowest in every category. U-Net++ achieved a Mean IoU of 0.8860 and Dice coefficient of 0.9372, representing improvements of 4.4% and 2.6% respectively over the U-Net baseline, and 11.8% and 7.2% over DeepLabV3+. According to the clinical Dice thresholds established by Zijdenbos et al. [44], a Dice score above 0.90 indicates near-expert agreement—U-Net++ and U-Net both exceed this threshold at the overall level, while DeepLabV3+ at 0.8740 falls within the clinically acceptable range of 0.70 to 0.90.

Metric	U-Net	DeepLabV3+	U-Net++
Mean Pixel Accuracy	98.66%	98.31%	98.99%
Mean IoU	0.8488	0.7929	0.8860
Mean Dice	0.9136	0.8740	0.9372
Sensitivity	0.9537	0.9379	0.9670
Specificity	0.9952	0.9938	0.9962
Precision	0.8797	0.8259	0.9108

Table 11: Overall Metrics Comparison

U-Net++ achieved a precision of 0.9108 compared to 0.8797 for U-Net and 0.8259 for DeepLabV3+, indicating that U-Net++'s dense skip connections produce substantially fewer false positive predictions at layer boundaries. This is consistent with the architectural expectation that nested encoder-decoder pathways reduce the semantic gap between encoder and decoder feature maps, leading to more precise boundary localisation.

7.4.2 SC Specific Performance

Since the main goal of this project is SC segmentation, the per class SC results in Table 12 are the most clinically meaningful comparison. U-Net++ achieved the highest SC IoU of 0.791 and Dice of 0.883, followed by U-Net with 0.738 and 0.849, and DeepLabV3+ with 0.650 and 0.788. All three models show a consistent pattern of high sensitivity but lower

precision for the SC class, thus indicating a tendency toward oversegmentation of the SC boundary rather than under detection. This is an expected characteristic of thin layer segmentation where boundary ambiguity causes models to expand predictions into adjacent tissue to avoid missing the target structure.

Metric	U-Net	DeepLabV3+	U-Net++
IoU	0.738	0.650	0.791
Dice	0.849	0.788	0.883
Sensitivity	0.929	0.899	0.944
Specificity	0.997	0.995	0.998
Precision	0.781	0.701	0.829

Table 12: SC Class Comparison

DeepLabV3+'s relatively poor SC performance despite its deeper ResNet-50 backbone and multi-scale ASPP module suggests that global contextual reasoning alone is insufficient for precise SC boundary delineation. The absence of strong symmetric skip connections means that fine spatial detail suppressed during encoding is not effectively recovered, resulting in the smoother and less precise boundary predictions observed in the qualitative analysis. U-Net's explicit skip connections provide better spatial recovery, and U-Net++'s dense nested connections further refine this, producing the sharpest and most accurate SC boundaries of the three architectures.

7.4.3 Computational Requirements

The computational comparison reveals an important trade-off. U-Net is the most efficient architecture with 24.4 million parameters, 31.51 GFLOPs, and a latency of 1600ms. DeepLabV3+ has a slightly higher parameter count at 26.7 million and 36.93 GFLOPs with a latency of 1844ms, yet achieves the lowest segmentation accuracy of the three, thus making it the least efficient architecture in terms of the accuracy-to-computation ratio. U-Net++ has a similar parameter count to DeepLabV3+ at 26.1 million, but requires significantly more computation at 73.91 GFLOPs and 3729ms latency due to its dense nested decoder structure. This computational cost is the main trade-off of U-Net++'s superior accuracy.

Model	Parameters	GFLOPs	Latency (ms)
U-Net	24,436,804	31.51	1600.24

DeepLabV3+	26,678,356	36.93	1844.73
U-Net++	26,079,044	73.91	3729.75

Table 13: Computational Requirements

It is worth noting that all latency measurements were taken on CPU, which exaggerates the absolute values. In a clinical deployment scenario using a GPU, all three models would operate considerably faster. The relative differences between models would however remain consistent, meaning U-Net++ would still require approximately twice the inference time of U-Net regardless of hardware.

7.4.4 Clinical Recommendation

Based on the combined analysis of segmentation accuracy and computational requirements, U-Net++ is recommended as the preferred model for SC segmentation in ultra-high-resolution OCT images where accuracy is the primary concern, particularly in research and pharmaceutical evaluation contexts where precise SC thickness measurement is critical. For scenarios where computational resources are limited or real-time processing is required, U-Net represents the best balance of accuracy and efficiency, achieving clinically acceptable performance at significantly lower computational cost. DeepLabV3+, despite its deeper backbone, is not recommended for this specific task due to its consistently lower boundary precision and disproportionate computational demand relative to its accuracy.

8 Discussions

8.1 Conclusions

This project successfully developed and evaluated three deep learning architectures—U-Net, DeepLabV3+, and U-Net++—for the automated segmentation of the Stratum Corneum and surrounding epidermal layers in ultra-high resolution OCT images of living skin. All basic and advanced project objectives were met, including dataset labelling, model implementation, hyperparameter tuning, and quantitative and qualitative evaluation under a controlled experimental framework.

U-Net achieved a strong baseline with a Mean IoU of 0.8488 and Dice coefficient of 0.9136, confirming that skip connections effectively preserve spatial boundary detail in low-contrast thin layer structures. DeepLabV3+, despite its deeper ResNet-50 backbone and multi-scale

ASPP module, consistently underperformed with a Mean IoU of 0.7929 and SC-specific IoU of 0.650, thus demonstrating that global contextual reasoning alone is not enough for precise SC boundary detection without strong spatial skip connections. U-Net++ delivered the strongest performance across every metric, achieving a Mean IoU of 0.8860 and Dice coefficient of 0.9372, placing it in the near-expert agreement range. Its dense nested skip connections and progressive feature refinement consistently produced the sharpest and most accurate SC boundaries of the three models, at the cost of approximately twice the inference time of U-Net.

Overall, this project shows that deep learning is a viable and effective approach to automated SC segmentation in OCT imaging, contributing towards reliable non-invasive tools for dermatological research and the objective evaluation of cosmetic and pharmaceutical formulations.

8.2 Future Work

This project has uncovered several insights and practical limitations that point toward future research direction.

The main limitation is the size and diversity of the dataset. With 212 original images labelled by a single annotator, the models are trained and tested on a relatively small ground truth that may not generalise across different skin types, anatomical sites and patient demographics. Future work should focus on the creation of a larger multi-annotator dataset with formal inter-observer agreement analysis, which would both improve the model robustness and produce more clinically accurate performance benchmarks.

An ablation study would provide deeper insight into the contribution of individual design choices to model performance. Specifically, investigating the effect of removing CLAHE preprocessing, replacing the hybrid loss function with standard cross-entropy, disabling data augmentation, and removing the squeeze-and-excitation attention modules in U-Net++ would quantify how much each component contributes to the final results. This analysis represents a natural and relatively straightforward extension of the existing experimental framework.

Looking further ahead, transformer-based segmentation models such as TransUNet and Swin-UNet have shown strong performance in medical image segmentation by combining convolutional local feature extraction with global self-attention mechanisms. These architectures may be particularly well suited to OCT segmentation where both fine boundary detail and broad contextual tissue organisation are important, and represent a natural progression beyond the CNN-based models tested in this project.

Ultimately, the ultimate validation of these models would come from comparing automated SC thickness measurements against established reference methods such as tape stripping gravimetry or confocal Raman spectroscopy. If deep learning segmentation can be shown to produce thickness estimates with these gold standard methods, it would establish OCT based segmentation as a viable non-invasive alternative for SC characterisation in both clinical dermatology and industrial testing, fulfilling the core motivation that drove this project.

8.3 Sustainability and Ethical Considerations

Every model trained, every hyperparameter tuned, and every failed experiment in this project consumed energy. This is rarely discussed in machine learning research, yet it is an important question: what is the true cost of building intelligent systems, and who bears it?

Training three deep learning models across 40 epochs each, including failed runs and hyperparameter tuning iterations, required an estimated 40 or more hours of GPU computation on Google Collaboratory's NVIDIA T4 infrastructure. While cloud computing removes the need for specific hardware, it does not eliminate the environmental impact the data centers powering these computations consume substantial electricity and water for cooling. The use of early stopping across all models helped reduce unnecessary computation by terminating training once validation performance plateaued, representing a small but meaningful step toward more efficient and responsible resource utilisation. Future work in AI should consider more efficient training strategies such as mixed precision training, neural model search, or knowledge distillation to achieve comparable performance at a reduced computational footprint.

Beyond energy consumption, the broader environmental and social implications of this project are largely positive. Automated SC segmentation has the potential to reduce reliance

on invasive skin biopsies in clinical assessment, contributing to less resource intensive diagnostic pathways. In the pharmaceutical and cosmetic industries, objective and reproducible thickness measurement tools could reduce the waste and costs associated with ineffective formulation testing, accelerating the development of more targeted products. These applications represent a genuine social and economic benefit, making dermatological analysis more accessible, less invasive, and more efficient at scale.

On the ethical side, the models developed in this project are powerful but complex. U-Net++ produces visually convincing segmentations, yet cannot explain why it placed a boundary where it did. In a research context, this is acceptable. Comparatively, in a clinical or pharmaceutical context a model that systematically overestimates SC thickness due to training data bias could lead to incorrect clinical conclusions or flawed product evaluations. Optimisation is not the same as understanding, and Dice score does not guarantee clinical reliability. The models developed here should therefore be treated as research instruments rather than clinical tools, and any real world deployment would require more independent validation across diverse populations, and anatomical sites before use.

A further ethical concern is annotation bias. All 212 images were labelled by a single annotator, without formal dermatological training. Model performance is therefore measured against one individual's interpretation of layer boundaries rather than verified clinical standard. This is a known limitation of a single annotator medical imaging datasets and means that reported accuracy figures, while strong, should be interpreted with appropriate caution.

Finally, the dataset used in this project was handled in full compliance with the University of Sheffield’s data protection guidelines. No new patient data was collected, no direct interaction with participants took place, and all images were stored securely on a protected university drive and encrypted storage devices. All data will be permanently deleted upon project completion. These measures ensure that the work meets the ethical standards required for academic research involving medical imaging data.

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